AI-INTEGRATED MULTI-MODAL ACADEMIC EVALUATION, PERFORMANCE INTELLIGENCE, AND AUTOMATED VIVA ASSESSMENT SYSTEM

INDIVIDUAL THESIS

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Automated Evaluation of Handwritten Entity Relationship Diagrams Using Object Detection, OCR, and Large Language Models.

R25-SE-025

B.Sc. (Hons) Degree in Information Technology

Specialized in Software Engineering

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Sri Lanka Institute of Information Technology

Sri Lanka

October 2026

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ABSTRACT

Reliable assessment of diagram-based examination answers remains difficult because handwritten Entity-Relationship (ER) diagrams combine textual labels with visual structures, and student submissions may contain ruled-paper lines, noise, overlaps, and varied drawing styles. This research investigates an image-based approach for extracting and evaluating handwritten ER diagrams from scanned or camera-captured answer sheets. The wider project proposal describes an AI-integrated academic assessment platform with performance intelligence and automated viva assessment; the completed implementation evidence supplied for this thesis concerns the Diagram Evaluation Engine for handwritten ER-diagram assessment.

The implemented pipeline combines a custom YOLOv8 Large object detector, optical character recognition (OCR), OpenCV-based line detection, structural reconstruction, and rubric-guided grading with a locally hosted large language model. The detector identifies five component classes: Attributes, Entities, Primary Keys, Relationships, and Weak Entities. Detected crops are processed through OCR.Space OCR Engine 3, with SHA-256-based caching of processed crops. Canny edge detection and probabilistic Hough line detection are used to estimate line segments, which are associated with the nearest detected objects to produce candidate connections. Detected items and connections are then represented in structured JSON and passed to a configurable marking rubric.

The model configuration uses YOLOv8l, image size 640, batch size 8, 100 training epochs, and an inference confidence threshold of 0.60. On a dataset containing 350 training images and five validation images, the reported preliminary results are 88.4% precision, 61.3% mAP@0.5, and an average inference time of 37.1 milliseconds per image. Because the validation set is extremely small, these values are treated as an initial feasibility check rather than evidence of generalisation. OCR accuracy, structural reconstruction accuracy, and agreement with expert grading remain unmeasured in the supplied results.

Keywords: Keywords: Automated assessment; handwritten diagram recognition; Entity-Relationship diagram; YOLOv8; OCR; structural reconstruction; rubric-based grading.

ACKNOWLEDGEMENTS

I would like to express my sincere gratitude to my supervisor and co-supervisor for their guidance, constructive feedback, and encouragement during the planning and development of this research project. Their advice has helped shape the research problem and strengthen the technical direction of the work.

I also acknowledge the collaboration of the research team members, whose participation in the broader research effort is represented in the related implementation paper. The exact allocation of individual technical tasks should be confirmed by the team and is intentionally not inferred in this thesis where the supplied documents do not state it.

I thank the Department of Information Technology and the Sri Lanka Institute of Information Technology for providing the academic environment in which this research was developed. I am grateful to lecturers, peers, and others who contributed discussion, feedback, or encouragement during the project.

Finally, I acknowledge the authors and researchers whose published work on assessment reliability, offline sketch parsing, handwritten diagram recognition, and automated diagram assessment informed the research background. Their work provided a basis for identifying the practical gap addressed by image-based evaluation of handwritten ER diagrams.

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LIST OF ABBREVIATIONS

Abbreviation	Full form / meaning
AI	Artificial Intelligence
API	Application Programming Interface
AP	Average Precision
CNN	Convolutional Neural Network
CV	Computer Vision
ER	Entity-Relationship
F1	Harmonic mean of precision and recall
GUI	Graphical User Interface
HDE	Handwritten Diagram Extraction
IoU	Intersection over Union
JSON	JavaScript Object Notation
LLM	Large Language Model
mAP	mean Average Precision
OCR	Optical Character Recognition
SDG	Sustainable Development Goal
SHA-256	Secure Hash Algorithm producing a 256-bit digest
UML	Unified Modeling Language
YOLO	You Only Look Once

1 INTRODUCTION

1.1 Overall project context

Assessment is a central part of higher education because examination results influence progression, qualification, and the feedback students receive about their learning. In computing programmes, students are frequently asked to express their understanding through models and diagrams, not only through prose or multiple-choice responses. Entity-Relationship (ER) diagrams are a common example: they represent entities, attributes, keys, and relationships in a visual language used to reason about database structure.

Marking an ER diagram requires the examiner to interpret both what is drawn and what it means. A correct answer may be represented using slightly different layouts or shapes, while an apparently neat diagram may contain semantic errors. Manual review can therefore be time-consuming, especially where a module contains a large cohort or multiple exam sittings. This thesis explores how computer vision, OCR, image processing, structured representations, and rubric-guided language models can assist with the task.

The project proposal identifies a broad AI-integrated academic evaluation platform, including performance intelligence and automated viva assessment. The implementation paper supplied for this thesis, however, reports preliminary experimental results only for a focused Diagram Evaluation Engine. This report keeps the broad proposal context visible while evaluating only the capabilities supported by implementation evidence.

Traditional educational assessment relies on human judgement. Human judgement is necessary for many open-ended tasks, but marks can vary with interpretation, fatigue, rubric use, and the order in which work is reviewed. Cassagneau-Francis and Wyness discuss the comparative merits and limitations of teacher assessment and external examinations, illustrating that assessment reliability is a broader educational question rather than a problem solved by one marking format [1]. Brimi reports variation among teachers marking the same English paper, reinforcing the value of transparent criteria and careful moderation [2].

Repetitive assessment creates an additional operational challenge. Erturk, van Tilburg, and Igou report that boredom during repeated marking can be associated with lower essay marks [3]. The result does not establish that all marking is unreliable, but it motivates tools that can make repeated, rule-governed checks more consistent and reduce repetitive workload.

Automating diagram assessment is not equivalent to automating a simple multiple-choice question. The system must identify symbols, extract handwriting, infer connections from spatial evidence, construct a meaningful representation, and decide whether that representation satisfies a marking rubric. Each stage can introduce errors, so the design must be transparent about uncertainty and evaluate components separately.

The primary users envisaged by the project proposal are lecturers and examiners who assess diagram-based answers, students whose work is evaluated, and institutions that manage examinations. A lecturer may benefit from a first-pass score, criterion-level feedback, and an annotated image that makes the system's interpretation inspectable. A student may benefit from faster and more consistent feedback, subject to institutional policy and human review. An institution may benefit from a more scalable workflow for processing scanned answer scripts.

The system is intended as decision support rather than an unquestionable replacement for academic judgement. A diagram may be ambiguous, incomplete, or use a valid notation variant that is not represented in the training data. A responsible workflow should therefore permit the examiner to inspect the original image, review the reconstructed representation, and override a score where appropriate.

The Quality Education goal is a relevant motivation because reliable and explainable feedback can support learning. Alignment with an education goal is a motivation, not proof of impact. Any claim that the system improves fairness, learning outcomes, or marking time must eventually be tested with users and independently measured data.

The wider project proposal is titled "AI-Integrated Multi-Modal Academic Evaluation, Performance Intelligence, and Automated Viva Assessment System". It frames assessment automation as a connected institutional workflow that may eventually

include handwritten diagram processing, text-answer analysis, performance intelligence, and viva assessment. The scope of this individual thesis is narrower: it investigates the diagram-evaluation component and reports only the implementation evidence available for that component. The wider features are retained as project context and future integration opportunities, not as validated outputs of the present study.

The intended academic value is to make the first pass over diagram answers more repeatable and inspectable. The system is designed to expose its intermediate interpretations instead of returning a mark alone. A lecturer should be able to examine detected components, OCR text, candidate connections, reconstructed structure, the marking rubric, and criterion-level feedback before deciding whether to accept or amend a score.

1.2 Individual component context

The practical input considered in this research is a diagram drawn by a student on an examination answer sheet and then scanned or photographed. Such an image differs from a digital diagram exported from a diagram editor. Paper texture, printed rulings, margin lines, shadows, camera angle, image blur, handwriting variation, and overlapping connectors may all appear in the image. The system must operate on pixels rather than receiving a ready-made graph, PlantUML description, or structured JSON document.

The difficulty is compounded by the relationship between components. Identifying an entity rectangle is useful, but assessment also requires determining which attributes belong to it and which relationships connect it to other entities. OCR can read a label incorrectly; a line detector can confuse a connector with a printed rule; a reconstruction algorithm can assign an endpoint to the wrong nearby object. A downstream grader can then receive an inaccurate representation even if its own reasoning is sound.

The rationale for the proposed solution is therefore to build a staged pipeline in which each intermediate output is stored and can be inspected. The current research paper combines object detection, OCR, OpenCV line detection, structural reconstruction,

and rubric-based grading. The work begins with a narrow, measurable component and defines separate evaluations for the complete system.

The intended value is not merely to produce an automated mark. A useful assessment assistant should show the detected components, recovered labels, candidate connections, reconstructed diagram, criterion-level scores, and feedback. These artifacts make the result more traceable than a single scalar score and allow an examiner to identify a failure stage.

From a technical perspective, the project investigates whether object detection can provide a practical entry point for recognizing handwritten ER diagrams without reconstructing pen strokes from temporal information that is unavailable in a static image. It also investigates whether a structured intermediary representation can connect image processing to rubric-based reasoning.

From a research perspective, the important question is whether the complete pipeline can generalize beyond a few sample images. A small feasibility result may show that a model runs and produces detections, but it cannot establish reliability under different handwriting styles, page layouts, or exam conditions. This report consequently treats current metrics as preliminary and specifies additional experiments needed before institutional use.

The individual component begins with a static raster image of a student ER diagram. Its key engineering challenge is to recover both visual objects and the relations among them without access to the order in which the student drew the strokes. The current paper describes object detection, crop-level OCR, line detection, nearest-object association, JSON reconstruction, and rubric-based LLM assessment as the processing sequence. The detector is the only stage for which quantitative results are reported in the supplied paper; other modules must therefore be described as implemented or prototyped according to the source, but not as quantitatively validated.

The system boundary excludes the full marking of all examination question types, speech or video analysis for viva examinations, institutional performance analytics, and a full production-grade examination management system. These items are

mentioned in the wider proposal but are outside the empirical focus of this individual thesis.

1.3 Problem statement

The research problem is defined as follows: how can a system extract the visual and semantic structure of a handwritten ER diagram from a real examination answer-sheet image and evaluate it against a configurable marking scheme without relying on an already digital diagram representation?

This problem has interrelated subproblems. First, the system must detect component regions despite variable handwriting and page quality. Second, it must recover labels from crops. Third, it must infer graph connectivity from spatial line evidence. Fourth, it must form a machine-readable ER representation. Finally, the representation must be scored against criteria in a way that is traceable and open to human review.

The principal research questions are: (RQ1) How effectively can a YOLOv8l model detect the five targeted ER-diagram component classes in the available image dataset? (RQ2) How accurately can OCR and line association recover labels and candidate connections? (RQ3) How faithfully can the reconstructed representation capture the student's intended ER structure? (RQ4) To what extent do automate criterion-level marks agree with expert grading? (RQ5) Which stages contribute most to end-to-end errors under realistic answer-sheet conditions? Only RQ1 has preliminary quantitative evidence in the supplied implementation paper.

In operational terms, the problem is to transform an unstructured handwritten ER-diagram image into a sufficiently accurate, machine-readable semantic representation that can be assessed against a marking rubric. An error at any earlier stage can propagate: a missed entity removes its labels from downstream processing; an OCR error can change a name; an incorrect line association can invent a relationship; and an incomplete structure can lead the grading model to award marks on incorrect evidence. A credible solution must consequently make intermediate states visible and measure each processing stage separately.

1.4 Research gap

The research gap is the lack of a sufficiently validated end-to-end method for assessing handwritten ER diagrams taken directly from scanned or camera-captured examination answer sheets. Existing work described in the supplied documents commonly uses digitally structured diagram inputs, controlled sketch datasets, or stroke-reconstruction strategies. Those assumptions do not fully capture paper examination conditions where rulings, margins, noise, and overlapping connectors can affect the visual evidence.

A second gap concerns the connection between perception and marking. Symbol detection alone does not create a correct ER model. The system must recover text, decide which lines connect which objects, construct a structured representation, and compare that representation with a marking rubric. A failure in any early stage can produce an incorrect grade even if the grading model follows its instructions correctly.

A third gap is evaluation completeness. The preliminary paper reports object-detection measures, but OCR accuracy, connection reconstruction accuracy, and agreement with expert marks remain ongoing work. The research therefore addresses both a technical integration gap and an evidence gap: it builds an initial pipeline and identifies the independent tests needed to determine whether it is dependable.

The response proposed in this thesis is a staged image-to-grade pipeline. A custom object detector finds the five supported component classes. OCR recovers component labels. OpenCV estimates line segments, which are associated with nearby detections to propose connections. The pipeline converts these observations into structured JSON and provides the resulting structure and an explicit marking rubric to a locally hosted LLM for criterion-based grading.

The system is designed to preserve the distinction between observed evidence and inferred meaning. The LLM is instructed not to invent absent components or connections, to provide one output for every rubric criterion, to respect maximum marks, and to assign zero when required information is missing or unverifiable. These constraints are intended to limit unsupported interpretation; they do not guarantee that every score generated will be correct.

The implementation currently demonstrates the object-detection component quantitatively. The remainder is described as implemented or prototyped in the research paper, but the supplied result set does not contain accuracy measurements for those stages. The final response to the gap is therefore preliminary feasibility and an evaluation framework, not a claim that a production-ready automated examiner has already been validated.

The gap is not simply the absence of AI grading. The specific unresolved need is an image-first, paper-oriented pipeline that recognizes ER-diagram components, extracts their labels, reconstructs relationships from a photographed or scanned page, and evaluates the resulting structure while accounting for valid semantic alternatives. This thesis addresses the feasibility of connecting these stages; it does not claim that the gap is fully resolved by the small preliminary detector experiment.

1.5 Aim and objectives

The main objective is to develop and preliminarily evaluate an image-based system that extracts handwritten ER-diagram components and supports rubric-guided evaluation of student diagrams submitted as scanned or photographed answer sheets.

Specific objectives are to: (1) identify and localize the five supported diagram component classes; (2) extract textual labels from detected crops using OCR; (3) detect line segments and associate candidate connections with component regions; (4) construct a structured representation of entities, attributes, relationships, and links; (5) compare the extracted representation with configurable marking criteria using a locally hosted language model; and (6) define independent measures for evaluating each stage and the complete pipeline.

The wider project proposal also discusses answer-sheet preprocessing, multimodal academic evaluation, performance analytics, and automated viva assessment. These features form the broader direction of the platform but are not treated as completed outcomes here because the implementation paper does not provide corresponding module-level results. The scope of this thesis is therefore deliberately narrower than the full vision described in the proposal.

The aim of this individual work is to investigate and document a staged Diagram Evaluation Engine for handwritten ER-diagram answers, and to establish an evidence-led evaluation approach for determining whether the component can support academic marking. The specific objectives and their required evidence are summarized below.

Objective	Evidence required
Detect ER-diagram components	Precision, recall, F1, per-class AP, mAP, and qualitative error review
Extract text labels from component crops	Character accuracy, word accuracy, exact-label match, and OCR failure rate
Reconstruct candidate connections and ER structure	Connection precision/recall, entity/attribute association accuracy, and structural validity
Evaluate diagram structures with a rubric	Criterion-level agreement, mean absolute error against expert scores, and feedback review
Assess practical feasibility	End-to-end latency, failure rate, human review rate, and reproducible experiment records

Table 1.Research objectives and proposed evidence indicators

1.6 Research questions or hypotheses

The study is exploration and component focused. It does not claim to test a pre-registered causal hypothesis. Instead, it uses research questions to make the current feasibility work and the required follow-up evaluation explicit.

- RQ1: To what extent can a custom YOLOv8l detector identify the five selected ER-diagram component classes in the available labelled image dataset?
- RQ2: How accurately can OCR recover handwritten labels from detected crops under varied handwriting and image-quality conditions?
- RQ3: How reliably can line detection and endpoint association reconstruct candidate connections between recognized diagram components?
- RQ4: Can a structured ER representation support rubric-guided scoring that agrees sufficiently with expert assessment while accepting semantically equivalent valid diagrams?
- RQ5: What technical, ethical, and operational controls are required before the component could be used as lecturer's decision support?

Only RQ1 has preliminary quantitative evidence in the supplied implementation paper. RQ2-RQ5 remain open research questions because their required accuracy, agreement, and stakeholder measurements are not reported in the provided material.

1.7 Scope limitations and assumptions

The current implementation scope is handwritten ER-diagram component detection and a prototype path to structure reconstruction and rubric-based grading. Five classes are supported: Attributes, Entities, Primary Keys, Relationships, and Weak Entities. Inputs are raster images. Outputs described by the research paper include annotated

images, structured JSON, OCR outputs, detected connections, criterion-level marks, and feedback.

The scope excludes a claim of comprehensive recognition of all ER notations. Cardinality and participation constraints, multiple diagram notations, text-only answer grading, predictive student analytics, and multimodal viva evaluation are future extensions in the supplied material. The system also does not claim to identify the student or infer unobservable intent.

The methodological assumptions are that the input image contains one relevant diagram or a manageable diagram region; the trained detector uses a consistent class taxonomy; OCR results can be mapped back to detected crops; and the marking rubric can be expressed in explicit criteria and maximum scores. In practice, these assumptions need to be checked during validation. Images that are too blurred, incomplete, or out of distribution should trigger a review pathway rather than an unjustifiably confident grade.

The thesis assumes that the input image contains one relevant diagram region or that the detector can locate the relevant components within the image. It also assumes that a marking rubric can be expressed as explicit criteria with maximum marks and that detected objects can be assigned stable identifiers for later stages. These assumptions require testing. For example, multiple diagrams on one sheet, overlapping annotations, severely cropped components, or a rubric that leaves semantic decisions implicit may violate them.

The reported performance figures must be read with their evidence boundary: 350 training images and five validation images. The current material does not provide a separate independent test set, per-class values, confidence intervals, full pipeline latency, OCR accuracy, structural reconstruction scores, expert mark comparison, or usability-study results. These missing values are not inferred or fabricated in this thesis.

1.8 Individual contribution

The intended contributions are a practical image-processing pipeline for handwritten ER-diagram assessment, an explicit separation between symbol recognition and

connection reconstruction, an intermediate structured representation for grading, and a component-level evaluation plan. The preliminary detector experiment provides an initial feasibility signal and illustrates why small validation sets must be interpreted cautiously.

The research is not presented as a completed validation of all system components. Its contribution is both constructive and evaluative: it describes the current prototype and establishes the evidence required to determine whether the approach performs reliably on real examination papers.

Chapter 2 explains the research approach, architecture, dataset, model configuration, OCR, connection detection, structural representation, LLM grading, testing plan, and commercialization aspects. Chapter 3 reports the measurements available in the supplied implementation paper and discusses their significance and limitations. Chapter 4 concludes the thesis and outlines future work. References and appendices provide the source list, definitions, illustrative structures, reproducibility configuration, and recommended testing matrix.

1.9 Thesis structure

Chapter 1 introduces the wider project, this individual component, the problem, gap, aim, objectives, and scope. Chapter 2 reviews the domain foundations and related work. Chapter 3 describes research design, study materials, evaluation procedure, metrics, analysis approach, ethics, and reproducibility. Chapter 4 describes requirements, architecture, design choices, and professional practices. Chapter 5 documents the implementation, integration, test strategy, and deployment considerations. Chapter 6 reports the results supported by the available evidence and separates them from unevaluated functionality. Chapter 7 interprets the results and discusses contribution, limitations, and implications. Chapter 8 examines commercialization and sustainability. Chapter 9 concludes and sets out future research directions, followed by references, glossary, and appendices.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The literature review was scoped around four connected topics: reliability in human assessment; recognition of offline or handwritten diagrams from raster images; automated assessment and feedback for graphical representations; and the use of language models to evaluate structured diagram descriptions. The supplied proposal and implementation paper provide the starting bibliography. The review concentrates on the specific need to assess ER diagrams drawn on ordinary answer sheets rather than attempting a complete survey of computer vision, OCR, or educational measurement.

The literature was grouped by the primary input assumptions of each approach. Work that starts from online strokes differs from work that starts from a scanned image; work that grades diagram files exported by a digital editor differs from work that must first infer the diagram from pixels; and work that evaluates text-like structures differs from work that must recover relations between diagram objects. These distinctions are important because reported results across different datasets, output representations, and evaluation protocols cannot be treated as directly comparable numerical baselines.

The review is narrative rather than a registered systematic literature review. It does not claim exhaustive database coverage, formal risk-of-bias scoring, or a reproducible screening protocol. Its purpose is to explain the design rationale and research gap supported by the cited studies. The limited scope should be considered when interpreting the breadth of the conclusions.

2.2 Domain and theoretical foundations

An ER diagram is a conceptual data-model representation. Entities represent distinguishable object types or concepts; attributes describe their properties; primary keys identify entity instances; and relationships represent associations among entities. Weak entities depend on an identifying relationship or owner entity for their identification. ER notations differ in visual convention, so a system must not rely solely on one exact geometric appearance. It needs to identify visual components and then reason about their semantics and the relations among them.

Object detection predicts class labels and bounding boxes for objects in an image. For this project, the detector defines candidate regions for five component types. Precision measures the proportion of accepted predictions that match ground truth, while recall measures the proportion of ground-truth objects recovered. Average Precision summarizes a precision-recall curve; mean Average Precision averages AP over classes under a specified IoU threshold. These values depend on the annotation policy, confidence threshold, class distribution, split design, and evaluation implementation.

Optical character recognition converts the pixels in a crop into machine-readable text. Handwritten labels in ER diagrams may be small, slanted, abbreviated, or intersected by diagram lines. Preprocessing may improve contrast or binarization, but an aggressive transform can remove pen strokes or alter characters. OCR confidence and recognition output should therefore be retained as observations rather than silently treated as ground truth.

Connection reconstruction adds a spatial reasoning problem. The system must identify line evidence and decide which objects a line links. In the current prototype, Canny edge detection produces an edge map, and probabilistic Hough detection returns line segments. Matching line endpoints to nearby object boxes is a practical heuristic, but the nearest object is not necessarily the intended endpoint when objects are close together, a line crosses another line, or page rulings resemble diagram connectors.

Finally, rubric-based grading translates the reconstructed representation into criterion-level marks. A large language model may help interpret a structured description against a rubric, but it should be constrained to available evidence, return schema-valid output, respect mark bounds, and abstain or request review when input is ambiguous. Semantic equivalence matters because two diagrams may express the same database design while differing in layout. A grading module should assess meaning rather than pixel-level similarity where the rubric permits alternatives.

2.3 Related research

Research on assessment reliability provides the educational rationale for consistent criteria and transparent scoring. Cassagneau-Francis and Wyness examine the merits of teacher assessment in comparison with external examinations [1]. Their

work is relevant because it frames a tension between contextual teacher judgement and standardized measurement. Automated tools may help standardize selected checks, but they do not remove the need to design valid assessment criteria or to interpret unusual student answers carefully.

Brimi's study of grading high-school English work describes substantial variation when multiple teachers score the same paper [2]. Although the subject and setting differ from computer-science diagram marking, the finding illustrates how rubric and a scoring process can still leave room for disagreement. A diagram assessment system should therefore make each criterion explicit and define what evidence is required for marks.

Erturk and colleagues investigate how repetitive marking and boredom can be associated with lower marks [3]. The study motivates careful workload design, but it should not be overgeneralized into a claim that automated marking is inherently fair. An automated system can be consistently wrong if its visual extraction is inaccurate or its rubric is incomplete. Reliability must be assessed empirically by comparing results with expert judgements and by testing different types of answers.

Offline sketch recognition analyses a static image after drawing has finished. It does not have access to the order, speed, pressure, or timing of the strokes that would be available from a tablet-based online system. Wu and colleagues propose offline sketch parsing through sharpness estimation, reducing the number of candidate stroke groups considered by later recognition stages [6]. Their work demonstrates the value of domain-specific visual features and efficient candidate selection.

However, sketch parsing and examination-paper assessment are not identical tasks. An exam answer sheet may include printed rules and margins that were not part of the student's drawing. A line that touches or follows a printed rule may be difficult to distinguish from a connector. Handwriting, scanner quality, and overlap between symbols may also differ from the conditions used to evaluate a general sketch recognizer.

The present approach uses object detection on the raster image and then estimates connectivity with line detection. This avoids reliance on reconstructing the original

sequence of strokes, although it introduces its own limitations: an object detector localises component regions but does not itself determine the graph structure. The literature therefore supports a modular design while highlighting the need to test it on realistic exam-paper images.

Bresler, Pruša, and Hlaváč study recognition of offline flowcharts by reconstructing strokes and using online-recognition techniques [7]. The method is relevant because it attempts to bridge a static image and techniques designed for temporally ordered strokes. Such approaches can be valuable where line structure and drawing order are otherwise lost.

The limitation identified in the project materials is that stroke reconstruction may fragment a line, join unrelated segments, or fail at intersections and overlaps. Errors can be particularly consequential in diagrams, where a small change to a connector can alter the perceived relationship between two symbols. Noise and real examination-paper backgrounds make these cases more likely.

The proposed thesis workflow instead identifies object regions and then derives candidate connections using Canny edge detection and probabilistic Hough line detection. This direct image-processing approach is simpler to inspect, but nearest-object endpoint assignment may fail when several symbols are close together. The comparison is not a claim that one family of methods is universally superior; rather, it explains why the proposed method is suitable as a prototype and why connection accuracy requires its own evaluation.

Schäfer, Keuper, and Stuckenschmidt propose Arrow R-CNN for offline handwritten diagram recognition [8]. Their approach extends region-based object detection with arrow-head and arrow-tail key point prediction and diagram-aware post-processing. This is important because diagram recognition must reconstruct not only the symbols but also the structure that connects them.

The supplied research paper distinguishes symbol detection from structural understanding. A model such as YOLOv8 can locate likely entities, attributes, primary keys, relationships, and weak entities. These detections are a necessary input for later

stages, but the detector does not automatically establish which attribute belongs to which entity or the meaning of a connecting line.

Arrow R-CNN represents a more integrated treatment of symbols and connectors than the current prototype. The present work uses YOLOv8l for symbols and OpenCV-based line detection for connections. This separation makes each component easier to implement and inspect, but it can also create error propagation between stages. The literature suggests future options such as key point prediction, graph-based spatial reasoning, or a learned connector detector if sufficient labelled data becomes available.

Correia, Leal, and Paiva describe techniques for improving feedback in automated diagram assessment by comparing graph representations and generating formative hints [9]. Their work is relevant to the reporting stage because effective feedback should identify the nature of a structural mismatch, not simply label an answer wrong. The supplied project materials note that such digital diagram approaches require students to construct diagrams within a software environment rather than on paper examination scripts.

Ramachandran and colleagues investigated LLM-assisted evaluation of ER and schema diagrams created in a graphical environment and represented in a structured form [5]. Their approach demonstrates how language models can be used after diagram structures have been extracted. However, structured input does not solve the separate visual problem of locating handwritten shapes, recovering labels, and interpreting paper-based connectors.

Bouali and colleagues compare LLM-generated scores with teaching-assistant scores for UML class diagrams, using digital diagrams converted into textual descriptions [4]. This is relevant to the grading stage, but it does not demonstrate image-based recognition of handwritten exam answers. The distinction defines the central opportunity for the present study: combine the visual extraction problem with structured semantic comparison rather than assume that a diagram is already available in machine-readable form.

Together, the reviewed studies suggest that the task combines two families of challenge: recognizing diagram objects in static images and reasoning over

relationships or semantic correctness. Recognition systems can recover symbols, but automated educational feedback requires a representation that is meaningful in the target subject domain. Conversely, a language model may be able to assess a structured description but cannot repair missing or incorrect information unless that information is extracted from the image first.

2.4 Existing systems products and techniques

The reviewed studies cover assessment reliability, offline sketch parsing, stroke reconstruction, joint symbol-and-arrow recognition, and automated feedback or LLM-based grading. They show that individual parts of the problem have been studied, but the project's target scenario combines several conditions: the source is a static handwritten diagram, the page may contain printed lines or visual noise, and the final output must be a criterion-level grade grounded in a marking scheme.

The following table summarizes how the cited studies inform the design. It is a qualitative comparison based on the descriptions in the supplied proposal and paper, not a new systematic review or an independently re-evaluated benchmark.

The most relevant comparison is based on input format and output assumptions rather than a raw metric leaderboard. The table below synthesizes the approaches cited in the source documents. "Digital/structured input" indicates that the cited workflow depends on pre-existing digital objects, a GUI export, or a structured textual representation; it does not imply that the method itself lacks technical merit. "Paper-oriented input" describes the target of the present thesis, not a claim that the existing works cannot be adapted.

Study	Input / focus	Relevance and distinction
Cassagneau-Francis & Wyness [1]	Teacher assessment versus external examinations	Provides the wider educational assessment context; not a diagram-recognition method.
Brimi [2]	Variation in English-paper grading	Motivates transparent scoring criteria and moderation.
Erturk et al. [3]	Repeated marking and boredom	Provides evidence that repetitive grading can be associated with changes in marks.
Wu et al. [6]	Offline sketch parsing	Directly relevant to static images without temporal stroke data.
Bresler et al. [7]	Offline flowchart stroke reconstruction	Illustrates a reconstruction-based alternative and its dependency on inferred strokes.

Study	Input / focus	Relevance and distinction
Schäfer et al. [8]	Arrow R-CNN for handwritten diagrams	Shows object-based handwritten diagram recognition with attention to arrows/connectors.
Correia et al. [9]	Automated feedback for digital diagrams	Relevant to formative feedback, but the cited system uses digitally authored diagrams.
Bouali et al. [4]	LLM grading of digital UML class diagrams	Provides evidence for comparing LLM scores with teaching-assistant marks on structured descriptions.
Ramachandran et al. [5]	LLM evaluation of ER/schema diagrams	Relevant to semantic assessment, while the cited workflow uses digitally created/structured diagrams.

Table 2.Summary of related research and relevance

Commercial computer-vision platforms, hosted OCR APIs, and locally hosted language models offer different trade-offs. Hosted services can reduce infrastructure work but introduce connectivity, data-transfer, recurring-cost, and retention considerations. Local models can improve control over where processing occurs, but require hardware resources, model management, and evaluation of their own outputs. This thesis describes OCR.Space and a locally hosted LLM because those are the components named in the supplied implementation paper; it does not claim that either is universally superior to alternative products.

2.5 Gap synthesis and conceptual framework

The synthesis of the literature produces a concrete pipeline requirement. First, the system needs to locate visual components from static images. Second, it must extract label text and retain uncertainty. Third, it must infer candidate connections from geometry. Fourth, it must produce a consistent structural representation. Fifth, it must grade that representation against explicit rubric while allowing semantically valid alternatives. Each stage needs independent evaluation and inspectable outputs because failure can propagate to later stages.

The current component responds to this need by linking a YOLOv8 detector to crop-level OCR, OpenCV line detection, structured JSON reconstruction, and rubric-guided LLM scoring. The contribution is primarily architectural and feasibility-oriented at this stage. It is not yet evidence that every stage meets the required accuracy threshold or that automatic scores agree with subject experts.

The conceptual framework distinguishes the implemented/prototyped diagram-evaluation path from broader proposal items such as performance intelligence and

audio/video viva assessment. Those broader items are shown as future integration context to avoid confusing the full group proposal with the empirical boundary of this individual thesis.

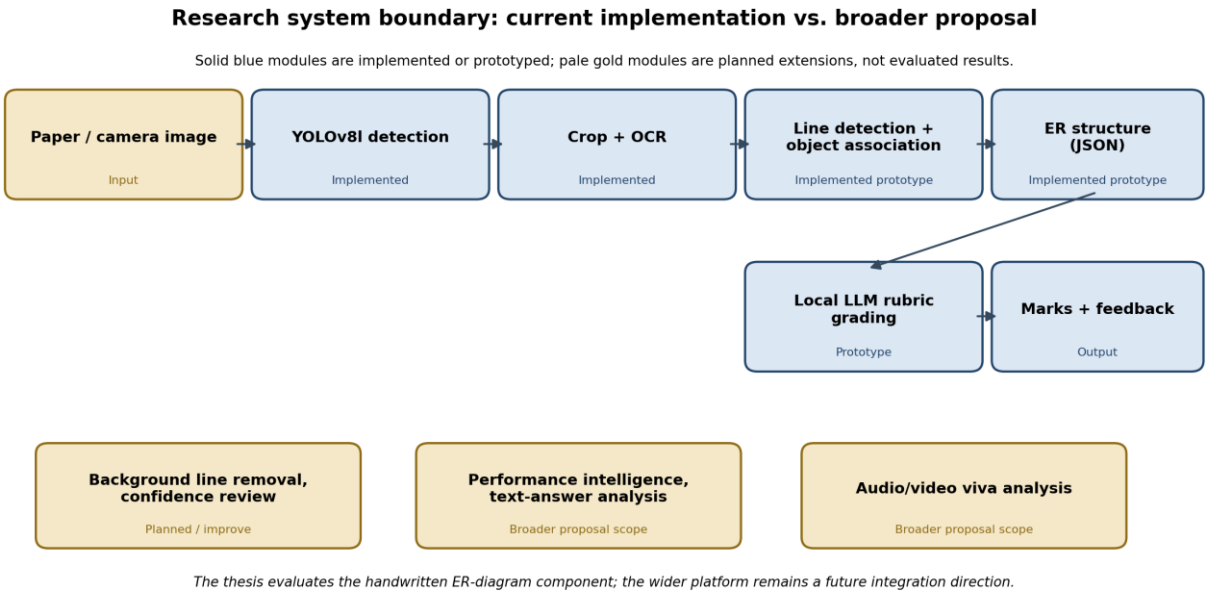


Figure 1.Research system boundary: current component and wider project.

3 RESEARCH METHODOLOGY

3.1 Research design

This chapter describes the method used to develop the Diagram Evaluation Engine and the evaluation procedure required to test it. The process follows a staged architecture: image input, object detection, crop generation, OCR, line detection, connection association, ER-structure reconstruction, rubric-based grading, and result presentation. A stage-based design makes intermediate outputs inspectable and allows errors to be traced back to their origin.

The research combines design-and-build activity with empirical model evaluation. Development creates an artefact intended to address a practical assessment problem; evaluation measures the artefact against explicit criteria. The supplied evidence contains quantitative detection results only. Other components are explained using the implementation description, while their quantitative evaluation is kept in the proposed test plan until data is available.

The terminology used in this chapter distinguishes three statuses. “Implemented” means the supplied paper describes the component as part of the working pipeline; “preliminary result” means a numeric result is reported but has important limitations; and “planned/future” means the feature is part of the proposal or recommended extension but has not been supported by implementation evidence in the supplied files.

The study uses an applied engineering approach. The system is developed as a sequence of components, each transforming one representation into another. The input is a raster image of a handwritten ER diagram. The output is a machine-readable set of detections, labels, candidate connections, reconstructed structures, and rubric-level feedback. The detector is trained on labelled examples and evaluated on a held-out validation set; downstream stages require additional labelled evaluation data.

The method prioritises transparency over hiding all intermediate decisions inside one model. A detection record can store class, bounding box, confidence, identifier, and crop path. OCR output can be linked to a detection. A line segment can be recorded with coordinates and candidate endpoint objects. These linked records permit the team

to identify whether an incorrect mark originates from a missed symbol, a wrong label, or a reconstructed connection.

The principal units of analysis are the image, detected object, component label, candidate connection, reconstructed diagram, and final criterion-level grade. Different metrics operate at different units. For example, object-detection precision is not a measure of final grading correctness. The research therefore avoids collapsing all stages into a single accuracy number.

The research design contains three related activities: system construction, component evaluation, and end-to-end evaluation planning. System construction implements the vision and grading pipeline. Component evaluation measures the detector using standard object-detection metrics. End-to-end evaluation is defined through OCR, connectivity, structure, and grading measures but is not complete in the supplied results.

The boundary of the evaluated system is important. The broader proposal describes an AI-integrated multi-modal platform for written answers, diagram answers, performance intelligence, and viva assessment. The implementation paper provides technical details for the handwritten ER-diagram component. Consequently, this thesis uses the broad project title to identify the research project, but its claims of implementation and measurement are restricted to the Diagram Evaluation Engine.

The research design combines design-and-development work with a preliminary quantitative experiment on the object-detection component. The engineering work specifies a pipeline, integrates a detector and downstream processing modules, and generates intermediate artefacts. The experiment then records detector metrics on the available train/validation data. Because the validation sample is very small and the downstream stages have no reported accuracy results, the overall study is best described as an early feasibility investigation rather than a completed validation of a grading product.

The unit of analysis for detector evaluation is an annotated object instance within a validation image; the runtime measure is reported per image. Future OCR analyses should use individual labels or character strings as the units of analysis, while

connection and structural analyses should use annotated object pairs, graph edges, or complete diagram structures. The grading analysis should use criterion-level and total scores for submissions independently marked by experts.

3.2 Data participants or study materials

The proposal planned to use handwritten ER diagrams from past examination answer sheets, digitized by scanner or camera. The implementation paper reports a custom labelled dataset of 350 training images and five validation images. The diagrams vary in layout, component positioning, and labels. The five detection classes are Attributes, Entities, Primary Keys, Relationships, and Weak Entities.

For object detection, annotation is required at component level. Each visible component should be assigned the correct class and a bounding box that encloses the region according to consistent annotation rules. Annotation guidelines should define how to label partially obscured objects, shapes touching a line, double-bordered weak entities, key attributes, and components whose boundaries are ambiguous.

The reported dataset is small, particularly the five-image validation set. It may not cover differences in handwriting, paper ruling, scan resolution, camera perspective, or component density. Before drawing conclusions about generalization, the project should expand the dataset and maintain a genuinely independent held-out test set. Training, validation, and test partitions should avoid near-duplicate images or crops from the same original diagram leaking into multiple partitions.

A repeatable annotation protocol is necessary because the meaning of precision and recall depends on consistent ground truth. The protocol should provide visual examples for each class and rules for boundary placement. Annotators should record uncertain cases instead of silently forcing ambiguous objects into a class. A second annotator or an adjudication pass can be used to estimate inter-annotator consistency.

Data quality checks should examine missing boxes, duplicate boxes, invalid coordinates, class-index mismatches, empty images, and disproportionate representation of one class. The dataset summary should include the number of images, number of instances per class, image-source category, and relevant image-quality characteristics where collection conditions permit.

Where answer sheets contain real student work, the research team should obtain the permissions required by the institution and remove personal identifiers not needed for the experiment. Dataset documentation should state whether samples are anonymized, whether they were collected from prior papers or created for research, and what restrictions apply to reuse. The supplied files do not provide a full dataset inventory or consent protocol; those details must be completed from the actual data-collection records.

The reported experiment used a custom labelled image dataset consisting of 350 training images and five validation images. The available paper does not provide the number of annotated instances per class, the precise source of every image, annotator agreement, split-selection procedure, or a separate held-out test set. These omissions restrict reproducibility and must be remedied before making claims about performance on new cohorts or examination settings.

No participant survey, lecturer interview, user study, or expert-grade comparison is reported in the supplied material. Therefore, the term "study materials" refers to labelled diagram images, annotations, detector outputs, OCR crops, candidate line segments, structural JSON, grading rubrics, and the preliminary detector result record. Any later use of real student answer sheets should follow the institution's approved data-access and anonymization processes.

3.3 Experimental or study procedure

The input is a scanned or photographed page or diagram image. Before inference, a practical system should validate image format, dimensions, orientation, and readability. Camera-captured images may require rotation correction, perspective normalization, and cropping. Extremely small or blurred images should be flagged because resizing cannot restore information that was never captured.

The proposal explicitly identifies paper rulings, rulers, margins, and background noise as challenges and describes removing or ignoring these structures. However, the implementation paper's confirmed pipeline describes YOLOv8 detection, crop preprocessing for OCR, and OpenCV line detection; it does not report a quantified ruler-removal experiment. This thesis therefore treats robust answer-sheet background

removal as a requirement and improvement direction rather than claiming that it has been solved.

Preprocessing needs care because a printed rule may overlap with a student's connector or component boundary. Aggressive line removal can delete genuine diagram evidence. A safer approach is to combine line orientation, repetition, spacing, proximity to detected objects, and later structural checks. Original images should be preserved so that all transformations remain traceable and can be reviewed.

A future preprocessing module should compare the estimated spacing and orientation of repeated paper lines, distinguish them from short diagram segments where possible, and retain a reversible transformation record. Deskewing should be evaluated on slanted pages, while contrast enhancement and denoising should be tested on both faint pencil and dark pen marks. Any preprocessing method should be measured against a manually reviewed set of original and transformed images.

Potential tests include clean photocopies, camera images with shadows, ruled paper, diagonally photographed sheets, low contrast handwriting, and diagrams whose connectors follow paper rules. The evaluation should count both beneficial removals and accidental removals of valid diagram strokes. A pipeline that removes background lines but also deletes many student connectors can reduce rather than improve the reconstruction quality.

The current prototype can accept a raster image and perform its reported detector and processing stages. To evaluate the full problem stated in the proposal, background-preprocessing requirements should be made explicit in the acceptance criteria and measured independently. The absence of reported results in this area is a limitation of the current evidence, not an indication that the issue is unimportant.

The detection model uses the Ultralytics YOLOv8 framework, with the large variant YOLOv8l selected for the reported experiment [10]. The custom model is configured for five component classes: Attributes, Entities, Primary Keys, Relationships, and Weak Entities. The detector processes an image and returns candidate class labels, bounding boxes, and confidence scores. These detections support later crop creation and spatial analysis.

Object detection is appropriate because the system must locate many component instances within one diagram image. A detection model can tolerate variation in position and scale more effectively than a rigid template-based detector, provided the training data covers the expected range. However, the model only predicts visual classes and locations; it does not prove that the detected object is semantically correct or structurally connected to another object.

The confidence threshold reported for inference is 0.60. The value controls the trade-off between accepting more candidate detections and suppressing low-confidence detections. A threshold should ultimately be selected on a validation set using the relative costs of false positives and false negatives. The reported threshold is a configuration value, not proof that 0.60 is optimal.

The implementation paper reports a YOLOv8l model trained for 100 epochs with an image size of 640 pixels and a batch size of 8. Google Colab was used for GPU-based training. The custom model was trained on the prepared labelled dataset and subsequently applied to previously unseen validation images. The results included precision, mAP@0.5, and average inference time.

These settings form the minimum reproducibility record that can be extracted from the supplied paper. A complete experimental record should additionally capture the exact library version, model checkpoint, random seed, training/validation split, optimizer settings, learning-rate schedule, augmentation settings, GPU type, and commit or experiment identifier. These details are not all specified in the source documents and are therefore listed as recommended supplementary records rather than asserted facts.

The choice of a large model may improve feature capacity but also requires more memory and compute than a smaller variant. For practical deployment, a controlled comparison of YOLOv8n/s/m/l or other appropriate alternatives should measure not only detection metrics but also latency, throughput, memory use, and the cost of false detections in later stages.

After training, the detector is applied to an input diagram image. Each accepted detection is recorded with a class label, bounding box, confidence score, identifier, and

crop path. The crop is then available for OCR. This creates a traceable mapping between the full image and its component-level evidence.

Bounding boxes require coordinate discipline. If an image is resized before inference, the system must preserve the scale factor to map detections back to the original image. Crops should be clipped to image boundaries and given sufficient padding to preserve handwritten characters near a shape border. A crop-generation bug can cause OCR errors even when the detector has localized the correct component.

The detector output should be reviewed for duplicate boxes, missed components, boxes covering multiple symbols, and low-confidence instances. Non-maximum suppression and confidence thresholds influence these outcomes, but their actual settings beyond the reported threshold are not fully specified in the supplied document. The implementation record should include any custom post-processing applied to model outputs.

The OCR stage reads labels from the detected crops. The implementation paper states that the crop is preprocessed to improve the quality of small labels and submitted to OCR.Space using OCR Engine 3 [11]. The returned text is associated with the corresponding detected object, and results are stored in JSON, CSV, and plain-text formats.

OCR is important because ER-diagram correctness depends on labels as well as shape geometry. If an entity name or attribute label is read incorrectly, the structural representation may be incomplete or may refer to the wrong concept. Small handwriting, cursive text, overlapping shapes, low contrast, and skew can reduce recognition quality. The OCR result should therefore preserve the raw prediction and, where available, confidence or service metadata.

Preprocessing can include scaling, grayscale conversion, contrast adjustment, thresholding, and noise reduction. Such transformations should be selected with actual evaluation samples rather than applied blindly. A threshold that helps dark ink can remove faint pencil strokes. The thesis does not report OCR accuracy, so no character- or word-level performance claim is made.

The OCR pipeline uses a SHA-256 digest of the processed image crop as a cache key. If the same processed crop is encountered again, the existing OCR result can be reused instead of making a repeated external request. This design can reduce duplicate API calls and can make repeated experiments more efficient when image crops are identical.

Cache design must ensure that the hash represents the actual OCR input. If preprocessing settings change, the processed crop hash should change accordingly; otherwise the cache might return a result generated from a different image representation. The cached record should also store the OCR engine or configuration identifier and the time the result was produced if those fields are available.

The use of an external OCR service introduces operational dependencies: network connectivity, service availability, API-key management, rate limits, and possible charges. API keys must not be embedded in public source code or written to diagnostic output. If the system is used with examination material, the team should confirm whether transmitting crops to an external service is compatible with the institution's privacy and data-handling requirements.

The connection-detection stage uses OpenCV to detect candidate lines in the original diagram image. Canny edge detection produces an edge map, and probabilistic Hough line detection identifies line segments. Each line is then considered as possible evidence of a connection between diagram components.

This step is needed because a collection of correctly detected shapes is not enough to reconstruct an ER diagram. The edges and relationships among objects determine the structure. Yet the image contains both meaningful connectors and irrelevant lines, such as paper rulings or object boundaries. The algorithm must distinguish line evidence that connects objects from line evidence that belongs to the page or to a shape.

The implementation paper reports that detected segments are matched to object regions through endpoint proximity. Duplicate connections are removed and lines whose endpoints map to the same object are ignored. The approach is described as a

practical approximation, and the paper explicitly notes that curved lines, crossing lines, disconnected lines, and poor endpoint placement can reduce accuracy.

For every line segment, the system considers its endpoints and searches for the nearest objects detected using bounding-box centers. When endpoints map to two distinct objects, a candidate connection can be stored using the identifiers of those objects and the line coordinates. Duplicate candidate edges can be removed by normalizing the object-pair ordering or comparing endpoint locations within a tolerance.

The nearest-center method is simple and computationally inexpensive, but its spatial assumption is fragile. If three objects lie close together, an endpoint may be assigned to the wrong object. A line may begin on the boundary of a symbol rather than near its center, and a single logical connector may be represented by multiple Hough segments. Crossings also create ambiguous intersections that should not automatically be treated as graph nodes.

Future improvements include estimating whether an endpoint touches a bounding-box boundary, using local edge orientation, checking for a visible connector path between the endpoint and object, combining collinear segments, and applying ER-specific constraints. A learned key point or connector detector could be considered if the project collects sufficient labelled examples. Whatever approach is selected, connection precision and recall must be measured against manually labelled links.

The structural reconstruction stage transforms component detections, OCR labels, and candidate connections into a structured representation. In the supplied implementation paper, Entities and Weak Entities are treated as entity types; Attributes and Primary Keys are treated as attribute types; and Relationships are represented as relationship components. Object identifiers and connection records link the component data.

A structured representation is necessary because a language model should not be asked to grade an ambiguous raw image without a controlled account of what the upstream modules observed. JSON makes the data explicit, machine-readable, and

suitable for validating field presence, types, and references. A reconstructed representation can also be compared directly with a reference model or marking rubric.

The schema should distinguish a detected fact from an uncertain interpretation. For instance, “a line segment between object_2 and object_7 was detected” is an observation; “attribute X belongs to entity Y” is a structural interpretation. Maintaining this distinction helps prevent uncertain connections from silently becoming definitive grading evidence. The supplied paper does not specify a formal schema version or a measured reconstruction accuracy, so the illustrative schema in Appendix A is not presented as a captured production output.

Before the grading stage, the reconstructed JSON should be validated. Each object identifier should be unique; every referenced connection endpoint should exist; coordinates should be numeric and within image bounds; the class label should belong to the supported taxonomy; and any criterion marks should be within the allowed range. Malformed JSON should be rejected or returned with a clear error rather than passed to the grader in an undefined state.

Structural validation can also flag suspicious relationships. An edge linked to the same object at both ends may be ignored under the reported approach; a candidate connection referencing a missing detection should be rejected; and an attribute with no associated entity can be represented as unresolved rather than assigned by guesswork. These checks do not replace semantic evaluation, but they reduce errors due to malformed data.

The reference solution should be represented using a comparable schema. Comparison rules need to account for labels, entity-attribute association, relationship structure, keys, and valid alternative layouts. If the task permits multiple valid diagrams, a strict pixel or coordinate comparison would be inappropriate. Graph structure and semantic equivalence offer a better basis, though their rules must be defined and tested with subject experts.

The grading module receives the reconstructed ER structure and a configurable marking guideline. The guideline contains criterion identifiers, descriptions, and maximum marks. The structured evidence and rubric are sent to a locally hosted large

language model. The output is expected to contain one result per criterion, assigned marks, and feedback.

The supplied implementation paper describes explicit constraints for grading: do not infer missing objects or connections, return one result for each rubric criterion, assign zero when required information is missing or unverifiable, respect maximum marks, and return structured JSON. These constraints are intended to limit unsupported assumptions and keep output aligned with the marking scheme.

An LLM output should not be assumed correct simply because it is valid JSON. Evaluation should check schema validity, arithmetic constraints, evidence support, feedback usefulness, and agreement with human expert marks. The model version, prompt template, decoding parameters, rubric version, and input structure should be recorded so that repeated runs can be analyzed. None of these controls are substituted for an expert comparison study.

The proposal identifies dynamic grading as a goal: the system should accept alternative diagrams that represent the same correct meaning rather than force students to reproduce one exact layout. This requirement is important because ER models can differ in placement, naming conventions, and some valid structural choices while preserving the intended data constraints.

A robust comparison should separate semantic criteria from presentation preferences. For example, correct identification of entities and key attributes can be checked independently from where those components appear on the page. Relationship meaning and cardinalities, where supported, should be evaluated as semantic constraints. A layout-only difference should not automatically lose marks if the rubric assesses database design correctness.

The current paper confirms a structured representation and rubric-guided grading approach, but the supplied evidence does not quantify how often alternative valid diagrams are accepted. Semantic equivalence is therefore a design objective requiring dedicated test cases created or reviewed by database design experts. Test cases should include equivalent diagrams with different layouts as well as near-miss diagrams that change the semantics.

The processing engine is described as Python-based and integrated with a web-server endpoint. The endpoint accepts an image and an optional evaluation guideline. It orchestrates object detection, crop generation, OCR, connection detection, reconstruction, grading, and result packaging. A defined API contract is necessary so that a web interface or other client can submit work and reliably interpret responses.

The output artifacts described in the paper include annotated images, structured JSON files, OCR results, candidate connections, criterion-level marks, and feedback. These outputs should share one evaluation identifier so that they can be retrieved as a set. Error responses should distinguish invalid inputs, OCR service failures, model failures, and grading failures rather than treating every failure as a low score.

For scalability, long-running evaluations may need asynchronous processing, job status, controlled concurrency, and persistent result storage. For security, uploaded files should be size-limited, validated, stored with least-privilege access, and removed according to an agreed retention policy. The supplied paper does not specify a production deployment architecture or service-level target, so these are implementation recommendations rather than claims about a deployed service.

The experimental procedure should be understood as two related tracks. The implementation track processes images through the pipeline and stores intermediate results. The measurement track compares selected outputs against manually verified ground truth. The supplied paper reports only the detector measurement track; the other stages are described technically but their accuracy evaluation remains future work. The following sequence therefore identifies both the processing stages and the evidence that should be captured for each one.

1. 1. Record the source image identifier, image dimensions, preprocessing configuration, and evaluation run identifier.
2. 2. Run YOLOv8l with the configured confidence threshold; record the class, bounding box, and confidence for each accepted detection.
3. 3. Crop the detected regions, preprocess the crops for OCR, and store each OCR result alongside its corresponding object identifier.
4. 4. Run edge and line detection on the diagram image; associate candidate line endpoints with nearby detected components and remove duplicates or self-links.

5. 5. Convert detections and candidate connections into a validated ER-oriented JSON representation; retain uncertainty rather than inventing unsupported edges.
6. 6. Submit the structured representation and versioned marking rubric to the grading module; validate the returned JSON, mark bounds, and per-criterion feedback.
7. 7. Compare the outputs with expert-labelled references for each stage and log failures, not only successful examples.

This staged procedure enables error attribution. For example, a missing attribute can be traced to a missing bounding box, an incorrectly cropped region, an OCR failure, an object-association error, or a reconstruction rule. Without the intermediate artifacts, downstream grade differences would be difficult to diagnose, and the system would be harder to trust.

3.4 Measures and evaluation metrics

Testing should cover component behavior, interfaces, and the end-to-end pipeline. Unit tests can validate coordinate conversion, crop bounds, JSON schema constraints, duplicate removal, and mark-range rules. Integration tests can verify that detector outputs reach OCR and reconstruction in the correct format, that external OCR failures are handled safely, and that the grading module returns the required criteria. System tests should use complete input images and compare the final result with an expert-reviewed reference.

Object-detection evaluation should report precision, recall, F1 score, per-class Average Precision, and mAP under a stated intersection-over-union threshold. OCR evaluation should measure character accuracy, word accuracy, exact label match, and failure rate. Structural evaluation should measure entity identification, attribute association, relationship identification, connection precision/recall, and unmatched connections. Grading evaluation should compare automated marks with expert marks using mean absolute error, criterion-level agreement, and suitable correlation or agreement statistics.

Inference latency for the detector must be distinguished from end-to-end latency. External OCR requests and language-model inference can dominate the total runtime. All reported measures should specify sample size, data split, environment, and

uncertainty. The current evidence only supports preliminary detector metrics, so the other measures remain an evaluation plan.

A rigorous evaluation should first freeze the model and processing configuration, then run it on a held-out dataset whose images were not used for training or model selection. Evaluation images should represent the likely deployment conditions, including different writers, paper layouts, image quality, component counts, and connector patterns. Dataset partitions should be documented before results are inspected.

The first evaluation pass should report metrics separately for each class. Next, OCR labels should be compared with ground truth for the same detected regions and, separately, for all expected components to account for missed detections. Connectivity should be evaluated against manually annotated object pairs. Reconstructed diagrams should be compared with a reference structure, allowing equivalent representations when justified by the rubric.

Finally, an expert marker or panel should score the same submissions using the approved marking scheme. Automated and human marks should be compared by criterion and total score, and disagreements should be categorized by root cause. A human-in-the-loop review should be tested on uncertain detections and low-confidence reconstruction cases. Any claim of faster marking should be supported by measured time-on-task against a suitable baseline.

Stage	Measure	Purpose
Object detection	Precision, recall, F1, per-class AP, mAP@0.5	Quantifies recognition and localization against annotated objects.
OCR	Character accuracy, word accuracy, exact-label match, failure rate	Measures label recovery quality on detected and expected regions.
Connection detection	Edge precision/recall, false-connection rate, endpoint assignment accuracy	Measures whether the right objects are connected.
Structure reconstruction	Entity identification, attribute association, relationship identification, graph validity	Measures semantic structure fidelity.

Stage	Measure	Purpose
Rubric grading	Mean absolute error, criterion agreement, score correlation, feedback consistency	Measures correspondence with expert marks and review quality.
System operation	End-to-end latency, failure rate, throughput, review rate	Measures the practical usability of the complete workflow.

Table 3.Proposed component-level and end-to-end evaluation measures

Metric definitions must be fixed before experiments. Detector runtime should be identified as model-only inference unless the timing includes image decoding, preprocessing, OCR network calls, line detection, structure construction, and LLM grading. A result from one stage cannot be used to infer the latency or accuracy of the complete system.

3.5 Data analysis

For the reported detector experiment, analysis is descriptive: the paper states the training/validation counts, configuration, precision, mAP@0.5, and average inference time. The available results do not include raw per-image predictions, confusion matrices, class-wise metrics, error counts, confidence intervals, or repeated runs. Consequently, inferential comparisons and robust estimates of generalizations cannot be calculated from the supplied summary alone.

For follow-up work, the recommended analysis is staged. First, report per-class and aggregate object detection metrics on a held-out test set. Second, compute OCR scores from exact ground-truth transcriptions and separately report OCR on detected regions versus all expected components to expose errors caused by missed detections. Third, compare predicted edges with manually annotated connection pairs and report false positives and false negatives. Fourth, compare reconstructed structures with canonical semantic representations, using an equivalence policy approved by ER-modelling experts. Finally, compare machine-generated rubric scores with blind expert marks at criterion and total-score level.

Subgroup analysis should be considered for handwriting style, image quality, paper rulings, image rotation, component density, and diagram complexity if the dataset contains sufficient examples. Subgroups with very small counts should be presented descriptively instead of making strong comparative claims. Error categories

should be inspected jointly with quantitative metrics because the same aggregate score can hide materially different failure patterns.

3.6 Ethics privacy security and responsible AI

Exam answer sheets can contain names, registration numbers, handwriting, and other personal information. Data should be collected and processed only with the required permissions, limited to what the research needs, and stored under control. Identifiers should be removed from images used for development wherever practical. Retention periods and deletion responsibilities should be agreed before any operational deployment.

The current OCR stage uses an external service and therefore may transmit crop images outside the local execution environment. The team must verify the service's current terms, data handling, and institutional acceptability. API keys should be kept in protected configuration, access logs should avoid storing sensitive content unnecessarily, and uploads should be validated for file type and size.

Methodologically, the small validation set limits generalization claims. The nearest-object connection algorithm has known failure modes, OCR quality can be poor for tiny or blurred labels, and LLM grading depends on upstream extraction quality. The proposal's background-line-removal requirement is not quantitatively evaluated in the current paper. These limitations should be made visible to users and addressed through further testing before high-stakes use.

Responsible deployment requires confidence-aware behavior. When detections are unclear, components overlap, labels cannot be read, or candidate connections are ambiguous, the system should mark the case for review rather than produce unwarranted certainty. Review decisions and overrides can be recorded to identify recurring failure modes and improve the model or rubric.

Fairness evaluation should examine performance across writing styles, diagram layouts, paper conditions, and student groups where lawful and ethically appropriate data is available. The aim is not to infer protected attributes, but to discover whether some types of valid submission are systematically more likely to be misread. An error

review process should allow students or markers to challenge a result under institutional procedures.

Security controls should include least-privilege access, secure transport, careful storage of uploaded material, input validation, abuse protection, dependency management, and logging that is useful without exposing answer content. For educational assessment, governance is part of engineering design: a system that is fast but impossible to inspect or appeal would not satisfy the practical need identified by the research.

Examination answer sheets are educational records and may contain names, registration numbers, handwriting, or other identifying information. Before collecting or processing real student work, the research team should establish an authorized purpose, permissions, retention period, access list, anonymization procedure, and deletion process. When an external OCR service is used, the data-transfer path and service terms should be reviewed; sensitive student data should not be uploaded without institutional approval.

Responsible use also requires a human-in-the-loop workflow. Marks that affect student progression should not be issued solely based on an unvalidated prototype. Low-confidence detections, uncertain graph connections, malformed structures, and disagreements between rubric evidence and model feedback should trigger review. Audit records should identify the model and rubric versions, preserve the evidence used to make a recommendation, and allow an examiner to correct or reject the automated result.

3.7 Reproducibility and research artefacts

The reported experiment used Google Colab for GPU-based training of the YOLOv8l model. The custom dataset included 350 training images and five validation images. The model was trained for 100 epochs with a 640-pixel image size and a batch size of eight, and the inference confidence threshold was set to 0.60. The paper reports average detector inference time of 37.1 milliseconds per image.

To reproduce these results, the experiment record should identify the dataset version and hashes, annotation files, train/validation split, configuration file, package

versions, saved model weights, and inference script. The computing environment should record the GPU and CUDA/runtime configuration. Results should be linked to a unique run identifier so that chart images and output files can be traced to a specific checkpoint.

The five-image validation split is too small for strong generalization claims. A single missed or correct detection can materially change the reported metrics when the sample is tiny. Results should therefore be treated as an initial sanity check and repeated on a larger independent test set before deployment decisions or comparative performance claims are made.

A complete paper-ready experiment should report the number of images and annotated instances per class, how the images were split, whether near duplicates were removed, and whether the test set was kept untouched during model selection. It should include the model checkpoint identifier, the exact validation command, and the definitions used for precision and mAP. Runtime should be measured under a stated environment and should distinguish model-only inference from full pipeline processing.

All intermediate files should be named or indexed in a consistent way. The original input image, model detections, OCR crop and output, line segments, reconstructed JSON, rubric, and grading output should be tied to one evaluation ID. This enables error analysis to follow a concrete case through the pipeline and supports transparent review of an unexpected result.

Reproducibility also requires documenting unsuccessful cases, not only illustrative successes. The results section should include representative false positives, false negatives, OCR failures, misconnected lines, malformed structures, and grading disagreements. Where privacy policy prevents publishing raw answer sheets, anonymized or synthetic examples can be used, provided that their status is stated accurately.

A reproducible release should preserve the dataset version and split manifest, annotation schema, class mapping, training configuration, code revision, model checkpoint identifier, package versions, inference command, hardware environment,

and raw metric outputs. Every image evaluation should have a stable evaluation identifier linking the original image to the annotations, detections, crops, OCR results, line segments, reconstructed JSON, rubric version, and grading response.

Where answer sheets cannot be distributed due to privacy or copyright restrictions, the thesis should identify the access limitation and provide non-sensitive artefacts such as annotation examples, synthetic diagrams, configuration files, metric definitions, pseudocode, and anonymized error summaries. Reproducibility does not require publication of personally identifying student data; it does require enough methodological detail for another researcher to understand and attempt to reproduce the analysis under appropriate permissions.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The component is intended to accept scanned or photographed ER-diagram images and return both machine-readable evaluation artefacts and human-inspectable evidence. Requirements below distinguish behavior described in the supplied paper from quality requirements that must be verified during further development.

ID	Requirement	Priority / verification note
FR-01	Accept an image of a handwritten ER diagram.	Core; validate image format, dimensions, and empty/corrupt files.
FR-02	Detect Entities, Weak Entities, Attributes, Primary Keys, and Relationships.	Implemented detector; evaluate on a larger held-out set.
FR-03	Generate crops and associate OCR text with object identifiers.	Described as implemented; OCR accuracy not reported.
FR-04	Detect candidate lines and associate endpoints with components.	Prototype described; connection metrics not reported.
FR-05	Construct a structured JSON representation of objects and candidate links.	Validate schema, identifiers, and references.
FR-06	Evaluate the structure against a configurable rubric and return criterion-level output.	Grading module described; expert agreement not reported.
FR-07	Display annotated images and intermediate outputs for review.	Reviewability requirement; validate with examiner users.
NFR-01	Protect student data and service credentials.	Access control, safe storage, and secret management required.
NFR-02	Return predictable errors when a stage fails.	Test invalid input, OCR/API failure, and invalid model output.
NFR-03	Provide reproducible results and audit information.	Record configuration, versions, evaluation identifiers, and logs.
NFR-04	Scale to examination batches without cross-record mixing.	Load and concurrency testing required.
NFR-05	Support human review of uncertain or high-impact cases.	Required before high-stakes operational use.

Table 4.Functional and non-functional requirements

The main acceptance principle is evidence traceability: every awarded mark should be connected to a rubric criterion and to the extracted structure that supports that decision. A user interface that displays only the final score would not satisfy the intended transparency goal.

4.2 Component architecture within the complete solution

The architecture links visual recognition with semantic evaluation. The first stage locates objects in an image using YOLOv8l. The detector's bounding boxes are used

to create crops for OCR and to provide spatial anchors for line association. OpenCV produces candidate line segments from the original diagram image. Endpoint association maps these segments to nearby detections, producing a candidate connectivity graph.

The detections, labels, and connections are then transformed into a structured ER representation. The grading module receives this representation along with a configurable rubric. Its output is structured JSON containing criterion identifiers, assigned marks, and explanatory feedback. The service also returns or stores the intermediate artifacts so that a user can inspect the system's interpretation.

The architecture is sequential, meaning that an upstream error can contaminate downstream stages. A false entity detection can create a spurious object; incorrect OCR may rename an attribute; a missed connector may remove a relationship; and the grading model may then apply its rubric correctly to incorrect input. This dependency motivates both stage-level testing and end-to-end testing.

Each stage produces artifacts that can support debugging and audit. Detection output contains object classes and bounding boxes. Crop files make OCR input inspectable. OCR results are stored in JSON, CSV, and plain-text formats. Line detection results include candidate segments and mapped endpoints. Reconstruction produces a structure representation that can be compared with a reference answer. Grading output includes criterion-level scores and feedback.

The pipeline should preserve stable identifiers across these artifacts. If a detected object has identifier obj_12, then the associated crop, OCR label, any line endpoint mapped to it, and the reconstructed component should reference that identifier. Stable identifiers prevent accidental mixing of evidence when a diagram has many components.

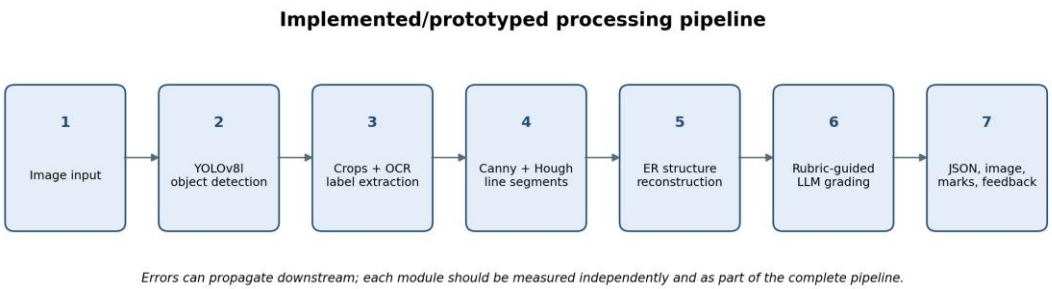


Figure 2.Processing pipeline for handwritten ER diagram evaluation.

The architecture is a pipeline rather than a single model. Object detection provides component regions; OCR adds textual labels; line detection provides candidate connectivity; reconstruction converts objects and links into a structured representation; and the rubric module evaluates the result. The final response should package the mark and feedback with the artefacts needed for review.

The broader R26-SE-025 proposal describes performance-intelligence and viva-assessment capabilities. These should connect to the ER-diagram component through explicit service contracts rather than being mixed into the detector itself. The current thesis treats those platform capabilities as future integration scope, because the supplied paper does not report their implementation or evaluation as part of this component.

4.3 Design alternatives and decisions

Three main alternatives informed the solution design. The first is reconstruction of the drawing strokes followed by online-style recognition. That approach can capture structure in some cases, but the original temporal stroke order is absent from a static scan, so strokes must be inferred and may be fragmented or incorrectly joined. The second is object-based recognition, in which a detector locates component instances directly in the raster image. The present implementation selects YOLOv8l for its training and inference workflow and uses a custom class set. The third is a hybrid approach that combines object detections with image-processing heuristics for line evidence and a semantic module for rubric scoring.

The chosen hybrid design separates tasks that require different evidence. The detector is responsible for object localization, not deciding the semantic correctness of the full ER model. OCR is attached to each object crop, rather than applied to the entire sheet without region context. OpenCV line detection provides candidate geometric links but does not by itself determine the semantic meaning of an ER relationship. A structured representation forms the interface to the grading stage so that grading logic does not need to interpret raw image pixels directly.

For OCR, the described implementation uses OCR.Space OCR Engine 3. A local OCR model could reduce external-service dependencies, while a hosted API can

reduce model-deployment work. The available evidence does not provide a controlled comparison, so the thesis records this as a project choice rather than an experimentally proven best choice. Similarly, locally hosted LLM grading is described by the supplied paper, but alternative models have not been compared in a controlled grading-agreement experiment.

For line association, nearest-object endpoint matching is computationally straightforward and easy to inspect. It is also vulnerable when objects are crowded or when a line endpoint is not clearly separated from the object border. More complex alternatives include connector key point detection, graph-based spatial reasoning, orientation-aware matching, and learned edge prediction. Those methods are plausible future improvements but require labelled connectivity examples to train or validate them.

4.4 Detailed design

The object detector returns a class label, confidence score, bounding-box coordinates, and an object identifier. Bounding boxes should be normalized to a documented coordinate convention, and crop generation should clamp coordinates to image bounds. A crop should preserve enough context for text recognition while avoiding excessive neighboring strokes. Detector output and crop paths should be stored using stable names so that a later analysis can identify exactly which region produced an OCR result.

The OCR module associates recognized text with the detection identifier. Preprocessing can include resizing, grayscale conversion, thresholding, denoising, or contrast adjustment; however, it should be applied conservatively and recorded as configuration. The SHA-256 cache described in the paper avoids repeating identical OCR requests for identical processed crops. A cache key should include preprocessing/version information when those settings affect the image content; otherwise, stale results may be reused after a processing change.

The connection module builds candidate line segments from Canny edges and probabilistic Hough detection. Candidate endpoints are matched to the nearest detected object using geometric distance to object centers or bounds. Duplicate candidate links

are removed, and self-links are ignored according to the described approach. The output should retain coordinates and confidence/uncertainty fields so a reviewer can inspect questionable cases. Crossings, curved lines, and background rulings are known problem cases for this heuristic.

The reconstruction module converts the set of components and candidate links into an ER-oriented JSON structure. Entities and weak entities are represented as entity types; attributes and primary keys as attribute types; and relationship objects as relationship types. The representation must validate that each referenced object exists, that each relationship follows allowed cardinality or endpoint rules where those rules are available, and that every component has a stable identifier. The present paper does not report a measured structural-validity rate or support for cardinalities and participation constraints.

The grading module receives the reconstructed structure and a rubric containing criterion identifiers, descriptions, and maximum marks. Its instructions constrain grading to the extracted evidence, require an entry for each criterion, prohibit marks beyond the stated limit, and assign zero where required evidence is missing or unverifiable. The result should be schema-validated before use. Semantic equivalence needs an explicit rubric policy so that layout changes do not automatically count as errors when the database meaning is the same.

The orchestration endpoint accepts an image and an optional evaluation guideline and coordinates these modules. The complete response is expected to include an annotated image, detection information, OCR results, candidate connections, reconstructed JSON, marks, and feedback. The source material does not specify a complete production API contract, authentication design, or service-level objective; these items must be established before institutional use.

4.5 Standards and professional practice

The thesis uses established software-engineering practices as design expectations: clear module boundaries, documented interfaces, input validation, structured error handling, versioned configuration, repeatable tests, and traceable changes. JSON artefacts should have a documented schema, including identifiers, coordinate units,

allowed class values, and rules for optional fields. A grading response should be treated as invalid until its schema and mark limits have been checked.

Experiments should define precision, recall, IoU, mAP, and runtime before evaluation. Document data splits to prevent near-duplicate leakage, retain an independent test set, and version the configuration, model, class definitions, and failure log.

Apply privacy-by-design and data minimization; explain marks using evidence and rubric criteria, and route uncertain cases for human review. Until validation is complete, treat the prototype as decision support, not autonomous marking.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The implementation paper reports a YOLOv8l model trained for 100 epochs with an image size of 640 pixels and a batch size of 8. Google Colab was used for GPU-based training. The custom model was trained on the prepared labelled dataset and subsequently applied to previously unseen validation images. The results included precision, mAP@0.5, and average inference time.

These settings form the minimum reproducibility record that can be extracted from the supplied paper. A complete experimental record should additionally capture the exact library version, model checkpoint, random seed, training/validation split, optimizer settings, learning-rate schedule, augmentation settings, GPU type, and commit or experiment identifier. These details are not all specified in the source documents and are therefore listed as recommended supplementary records rather than asserted facts.

The choice of a large model may improve feature capacity but also requires more memory and computing than a smaller variant. For practical deployment, a controlled comparison of YOLOv8n/s/m/l or other appropriate alternatives should measure not only detection metrics but also latency, throughput, memory use, and the cost of false detections in later stages.

The reported experiment used Google Colab for GPU-based training of the YOLOv8l model. The custom dataset included 350 training images and five validation images. The model was trained for 100 epochs with a 640-pixel image size and a batch size of eight, and the inference confidence threshold was set to 0.60. The paper reports average detector inference time of 37.1 milliseconds per image.

To reproduce these results, the experiment record should identify the dataset version and hashes, annotation files, train/validation split, configuration file, package versions, saved model weights, and inference script. The computing environment should record the GPU and CUDA/runtime configuration. Results should be linked to a unique run identifier so that chart images and output files can be traced to a specific checkpoint.

The five-image validation split is too small for strong generalization claims. A single missed or correct detection can materially change the reported metrics when the sample is tiny. Results should therefore be treated as an initial sanity check and repeated on a larger independent test set before deployment decisions or comparative performance claims are made.

The supplied implementation paper identifies a Python-based processing engine integrated with a web-server endpoint. YOLOv8 is used through the Ultralytics framework; OCR is requested from OCR.Space using OCR Engine 3; OpenCV provides edge and line detection; structured results are recorded as JSON and related formats; and rubric grading uses a locally hosted language model. Training is reported as having been conducted in Google Colab with GPU-based computation. Exact package versions, the GPU model, random seeds, model checkpoint hash, and deployment runtime are not stated in the provided report and should be added to the final experiment record.

Parameter	Reported value
Detector	YOLOv8 Large (YOLOv8l)
Image size	640 pixels
Training epochs	100
Batch size	8
Number of classes	5
Inference confidence threshold	0.60
Training environment	Google Colab with GPU-based computation
Training images	350
Validation images	5

Table 5.Model and experiment configuration reported in the paper

5.2 Component implementation

The detection model uses the Ultralytics YOLOv8 framework, with the large variant YOLOv8l selected for the reported experiment [10]. The custom model is configured for five component classes: Attributes, Entities, Primary Keys, Relationships, and Weak Entities. The detector processes an image and returns candidate class labels, bounding boxes, and confidence scores. These detections support later crop creation and spatial analysis.

Object detection is appropriate because the system must locate many component instances within one diagram image. A detection model can tolerate variation in position and scale more effectively than a rigid template-based detector, provided the training data covers the expected range. However, the model only predicts visual classes and locations; it does not prove that the detected object is semantically correct or structurally connected to another object.

The confidence threshold reported for inference is 0.60. The value controls the trade-off between accepting more candidate detections and suppressing low-confidence detections. A threshold should ultimately be selected on a validation set using the relative costs of false positives and false negatives. The reported threshold is a configuration value, not proof that 0.60 is optimal.

After training, the detector is applied to an input diagram image. Each accepted detection is recorded with a class label, bounding box, confidence score, identifier, and crop path. The crop is then available for OCR. This creates a traceable mapping between the full image and its component-level evidence.

Bounding boxes require coordinate discipline. If an image is resized before inference, the system must preserve the scale factor to map detections back to the original image. Crops should be clipped to image boundaries and given sufficient padding to preserve handwritten characters near a shape border. A crop-generation bug can cause OCR errors even when the detector has localized the correct component.

The detector output should be reviewed for duplicate boxes, missed components, boxes covering multiple symbols, and low-confidence instances. Non-maximum suppression and confidence thresholds influence these outcomes, but their actual settings beyond the reported threshold are not fully specified in the supplied document. The implementation record should include any custom post-processing applied to model outputs.

The OCR stage reads labels from the detected crops. The implementation paper states that the crop is preprocessed to improve the quality of small labels and submitted to OCR.Space using OCR Engine 3 [11]. The returned text is associated with the

corresponding detected object, and results are stored in JSON, CSV, and plain-text formats.

OCR is important because ER-diagram correctness depends on labels as well as shape geometry. If an entity name or attribute label is read incorrectly, the structural representation may be incomplete or may refer to the wrong concept. Small handwriting, cursive text, overlapping shapes, low contrast, and skew can reduce recognition quality. The OCR result should therefore preserve the raw prediction and, where available, confidence or service metadata.

Preprocessing can include scaling, grayscale conversion, contrast adjustment, thresholding, and noise reduction. Such transformations should be selected with actual evaluation samples rather than applied blindly. A threshold that helps dark ink can remove faint pencil strokes. The thesis does not report OCR accuracy, so no character- or word-level performance claim is made.

The OCR pipeline uses a SHA-256 digest of the processed image crop as a cache key. If the same processed crop is encountered again, the existing OCR result can be reused instead of making a repeated external request. This design can reduce duplicate API calls and can make repeated experiments more efficient when image crops are identical.

Cache design must ensure that the hash represents the actual OCR input. If preprocessing settings change, the processed crop hash should change accordingly; otherwise the cache might return a result generated from a different image representation. The cached record should also store the OCR engine or configuration identifier and the time the result was produced if those fields are available.

The use of an external OCR service introduces operational dependencies: network connectivity, service availability, API-key management, rate limits, and possible charges. API keys must not be embedded in public source code or written to diagnostic output. If the system is used with examination material, the team should confirm whether transmitting crops to an external service is compatible with the institution’s privacy and data-handling requirements.

The connection-detection stage uses OpenCV to detect candidate lines in the original diagram image. Canny edge detection produces an edge map, and probabilistic Hough line detection identifies line segments. Each line is then considered as possible evidence of a connection between diagram components.

This step is needed because a collection of correctly detected shapes is not enough to reconstruct an ER diagram. The edges and relationships among objects determine the structure. Yet the image contains both meaningful connectors and irrelevant lines, such as paper rulings or object boundaries. The algorithm must distinguish line evidence that connects objects from line evidence that belongs to the page or to a shape.

The implementation paper reports that detected segments are matched to object regions through endpoint proximity. Duplicate connections are removed and lines whose endpoints map to the same object are ignored. The approach is described as a practical approximation, and the paper explicitly notes that curved lines, crossing lines, disconnected lines, and poor endpoint placement can reduce accuracy.

For every line segment, the system considers its endpoints and searches for the nearest detected objects using bounding-box centers. When endpoints map to two distinct objects, a candidate connection can be stored using the identifiers of those objects and the line coordinates. Duplicate candidate edges can be removed by normalizing the object-pair ordering or comparing endpoint locations within a tolerance.

The nearest-center method is simple and computationally inexpensive, but its spatial assumption is fragile. If three objects lie close together, an endpoint may be assigned to the wrong object. A line may begin on the boundary of a symbol rather than near its center, and a single logical connector may be represented by multiple Hough segments. Crossings also create ambiguous intersections that should not automatically be treated as graph nodes.

Future improvements include estimating whether an endpoint touches a bounding-box boundary, using local edge orientation, checking for a visible connector path between the endpoint and object, combining collinear segments, and applying ER-

specific constraints. A learned key point or connector detector could be considered if the project collects sufficient labelled examples. Whatever approach is selected, connection precision and recall must be measured against manually labelled links.

The structural reconstruction stage transforms component detections, OCR labels, and candidate connections into a structured representation. In the supplied implementation paper, Entities and Weak Entities are treated as entity types; Attributes and Primary Keys are treated as attribute types; and Relationships are represented as relationship components. Object identifiers and connection records link the component data.

A structured representation is necessary because a language model should not be asked to grade an ambiguous raw image without a controlled account of what the upstream modules observed. JSON makes the data explicit, machine-readable, and suitable for validating field presence, types, and references. A reconstructed representation can also be compared directly with a reference model or marking rubric.

The schema should distinguish a detected fact from an uncertain interpretation. For instance, “a line segment between object_2 and object_7 was detected” is an observation; “attribute X belongs to entity Y” is a structural interpretation. Maintaining this distinction helps prevent uncertain connections from silently becoming definitive grading evidence. The supplied paper does not specify a formal schema version or a measured reconstruction accuracy, so the illustrative schema in Appendix A is not presented as a captured production output.

Before the grading stage, the reconstructed JSON should be validated. Each object identifier should be unique; every referenced connection endpoint should exist; coordinates should be numeric and within image bounds; the class label should belong to the supported taxonomy; and any criterion marks should be within the allowed range. Malformed JSON should be rejected or returned with a clear error rather than passed to the grader in an undefined state.

Structural validation can also flag suspicious relationships. An edge linked to the same object at both ends may be ignored under the reported approach; a candidate connection referencing a missing detection should be rejected; and an attribute with

no associated entity can be represented as unresolved rather than assigned by guesswork. These checks do not replace semantic evaluation, but they reduce errors due to malformed data.

The reference solution should be represented using a comparable schema. Comparison rules need to account for labels, entity-attribute association, relationship structure, keys, and valid alternative layouts. If the task permits multiple valid diagrams, a strict pixel or coordinate comparison would be inappropriate. Graph structure and semantic equivalence offer a better basis, though their rules must be defined and tested with subject experts.

The grading module receives the reconstructed ER structure and a configurable marking guideline. The guideline contains criterion identifiers, descriptions, and maximum marks. The structured evidence and rubric are sent to a locally hosted large language model. The output is expected to contain one result per criterion, assigned marks, and feedback.

The supplied implementation paper describes explicit constraints for grading: do not infer missing objects or connections, return one result for each rubric criterion, assign zero when required information is missing or unverifiable, respect maximum marks, and return structured JSON. These constraints are intended to limit unsupported assumptions and keep output aligned with the marking scheme.

An LLM output should not be assumed correct simply because it is valid JSON. Evaluation should check schema validity, arithmetic constraints, evidence support, feedback usefulness, and agreement with human expert marks. The model version, prompt template, decoding parameters, rubric version, and input structure should be recorded so that repeated runs can be analyzed. None of these controls are substituted for an expert comparison study.

The proposal identifies dynamic grading as a goal: the system should accept alternative diagrams that represent the same correct meaning rather than force students to reproduce one exact layout. This requirement is important because ER models can differ in placement, naming conventions, and some valid structural choices while preserving the intended data constraints.

A robust comparison should separate semantic criteria from presentation preferences. For example, correct identification of entities and key attributes can be checked independently from where those components appear on the page. Relationship meaning and cardinalities, where supported, should be evaluated as semantic constraints. A layout-only difference should not automatically lose marks if the rubric assesses database design correctness.

The current paper confirms a structured representation and rubric-guided grading approach, but the supplied evidence does not quantify how often alternative valid diagrams are accepted. Semantic equivalence is therefore a design objective requiring dedicated test cases created or reviewed by database design experts. Test cases should include equivalent diagrams with different layouts as well as near-miss diagrams that change the semantics.

The implementation is intentionally decomposed so each step can be executed and inspected independently. This is useful for troubleshooting, because an incorrect final score might originate in a detector miss, crop error, OCR substitution, connector-association error, incomplete JSON structure, or an unconstrained grading response. Keeping intermediate artefacts allows the team to classify and reproduce failures rather than treating the final grade as an opaque output.

5.3 Integration with the other components

Each stage produces artifacts that can support debugging and audit. Detection output contains object classes and bounding boxes. Crop files make OCR input inspectable. OCR results are stored in JSON, CSV, and plain-text formats. Line detection results include candidate segments and mapped endpoints. Reconstruction produces a structure representation that can be compared with a reference answer. Grading output includes criterion-level scores and feedback.

The pipeline should preserve stable identifiers across these artifacts. If a detected object has identifier obj_12, then the associated crop, OCR label, any line endpoint mapped to it, and the reconstructed component should reference that identifier. Stable identifiers prevent accidental mixing of evidence when a diagram has many components.

The processing engine is described as Python-based and integrated with a web-server endpoint. The endpoint accepts an image and an optional evaluation guideline. It orchestrates object detection, crop generation, OCR, connection detection, reconstruction, grading, and result packaging. A defined API contract is necessary so that a web interface or other client can submit work and reliably interpret responses.

The output artifacts described in the paper include annotated images, structured JSON files, OCR results, candidate connections, criterion-level marks, and feedback. These outputs should share one evaluation identifier so that they can be retrieved as a set. Error responses should distinguish invalid inputs, OCR service failures, model failures, and grading failures rather than treating every failure as a low score.

For scalability, long-running evaluations may need asynchronous processing, job status, controlled concurrency, and persistent result storage. For security, uploaded files should be size-limited, validated, stored with least-privilege access, and removed according to an agreed retention policy. The supplied paper does not specify a production deployment architecture or service-level target, so these are implementation recommendations rather than claims about a deployed service.

Component	Input	Output / artefact
Object detection	Full input image	Class, bounding box, confidence, ID, crop path
Crop generation and OCR	Detected crop	Preprocessed crop and recognized text per object
Connection detection	Original image and object boxes	Line coordinates and candidate endpoint object IDs
ER reconstruction	Object records, OCR labels, candidate links	Structured ER-oriented JSON
Rubric grading	Reconstructed JSON and rubric version	Criterion-level marks and explanatory feedback
Result packaging	All intermediate module outputs	Annotated image, JSON files, OCR results, marks, feedback

Table 6.Pipeline components, inputs, and outputs

Integration contracts should identify coordinate conventions, identifiers, field names, nullability, encoding, and error semantics. The OCR module should not need to infer an object identifier from a filename if the orchestrator can pass it explicitly. Similarly, the grading module should not re-interpret low-level bounding boxes; it should receive a validated semantic representation and the exact rubric version. Any failure should include the stage that failed, a machine-readable reason, and the partial outputs that remain safe to return.

The paper describes a pipeline and endpoint but does not provide measurable integration results, such as a success rate over a complete image batch, end-to-end latency, or an integration-test pass count. The presence of an integrated workflow should therefore not be treated as evidence of production reliability.

5.4 Verification and software testing

Testing should cover component behavior, interfaces, and the end-to-end pipeline. Unit tests can validate coordinate conversion, crop bounds, JSON schema constraints, duplicate removal, and mark-range rules. Integration tests can verify that detector outputs reach OCR and reconstruction in the correct format, that external OCR failures are handled safely, and that the grading module returns the required criteria. System tests should use complete input images and compare the final result with an expert-reviewed reference.

Object-detection evaluation should report precision, recall, F1 score, per-class Average Precision, and mAP under a stated intersection-over-union threshold. OCR evaluation should measure character accuracy, word accuracy, exact label match, and failure rate. Structural evaluation should measure entity identification, attribute association, relationship identification, connection precision/recall, and unmatched connections. Grading evaluation should compare automated marks with expert marks using mean absolute error, criterion-level agreement, and suitable correlation or agreement statistics.

Inference latency for the detector must be distinguished from end-to-end latency. External OCR requests and language-model inference can dominate the total runtime. All reported measures should specify sample size, data split, environment, and uncertainty. The current evidence only supports preliminary detector metrics, so the other measures remain an evaluation plan.

A rigorous evaluation should first freeze the model and processing configuration, then run it on a held-out dataset whose images were not used for training or model selection. Evaluation images should represent the likely deployment conditions, including different writers, paper layouts, image quality, component counts, and

connector patterns. Dataset partitions should be documented before results are inspected.

The first evaluation pass should report metrics separately for each class. Next, OCR labels should be compared with ground truth for the same detected regions and, separately, for all expected components to account for missed detections. Connectivity should be evaluated against manually annotated object pairs. Reconstructed diagrams should be compared with a reference structure, allowing equivalent representations when justified by the rubric.

Finally, an expert marker or panel should score the same submissions using the approved marking scheme. Automated and human marks should be compared by criterion and total score, and disagreements should be categorized by root cause. A human-in-the-loop review should be tested on uncertain detections and low-confidence reconstruction cases. Any claim of faster marking should be supported by measured time-on-task against a suitable baseline.

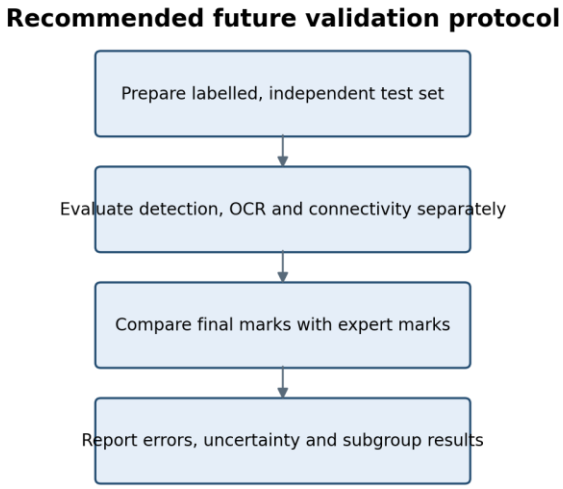


Figure 3.. Component-level validation workflow recommended for follow-up evaluation.

Verification should be performed at three levels. Unit tests should cover coordinate conversion, crop bounds, cache behaviour, duplicate-edge removal, JSON schema rules, and mark-range constraints. Integration tests should verify that detector output is passed correctly to OCR and reconstruction, that the application handles external OCR failures safely, and that the grading module returns all rubric criteria. System

tests should run complete images through the endpoint and compare the final structure and marks with expert-reviewed ground truth.

Test ID	Scenario	Expected behavior / evidence
T01	Clear diagram with all five supported classes	Detect components; report per-class precision/recall.
T02	Small or ambiguous handwritten labels	Return OCR output and flag uncertain text; measure exact-label match.
T03	Ruled answer sheet with horizontal lines	Do not treat every paper ruling as a connector; measure false edges.
T04	Crossing or overlapping connectors	Avoid unsupported junctions; measure connection precision/recall.
T05	Two objects in close proximity	Associate endpoints correctly or preserve an uncertainty flag.
T06	Partial/cropped symbol	Handle incomplete evidence without inventing an object.
T07	Semantically equivalent layout	Accept where the rubric policy permits; obtain expert review.
T08	Malformed grading JSON	Reject or recover safely; do not publish invalid marks.
T09	OCR service unavailable	Return a controlled error and preserve completed artefacts.
T10	Large batch of submissions	Maintain record isolation, throughput, and stable error rates.
T11	Low-confidence detections	Trigger human review were configured and measure review effectiveness.

Table 7.Recommended validation test matrix

The table is a proposed test plan, not evidence that these tests have passed. Actual execution records should include test date, image IDs, software and model versions, expected outcome, actual outcome, defect severity, and reviewer. A proper validation set should be larger and more representative than the current five-image set, and a final independent test set should be reserved for the final performance estimate.

5.5 Deployment and operation

The supplied paper describes the processing engine as integrated with a web-server endpoint that accepts an image and optional evaluation guideline. It does not document complete production deployment, authentication mechanism, queueing strategy, database schema, operational monitoring setup, or uptime measurements. This thesis therefore records endpoint integration as part of the described prototype while treating production operation as a set of required engineering tasks.

An operational deployment should validate MIME type and file contents, limit image size, normalize image orientation, apply request timeouts, and protect API keys and model endpoints. Long-running requests may require asynchronous jobs and job-

status polling. Batch processing should isolate each evaluation ID and ensure that outputs from one answer sheet cannot be associated with another. Logs should capture errors and timing without exposing student names or raw answer content unnecessarily.

Operational readiness also requires model and rubric versioning, health checks, metrics for failed requests, storage retention rules, and a documented recovery process. Before use during an actual examination, the institution should define how an examiner can override a result, how a disputed mark is reviewed, and who is accountable for the final grade. The current paper does not provide evidence that these production controls have been deployed or tested.

6 RESULTS AND VALIDATION

6.1 Research results

This chapter reports the experimental evidence available in the supplied implementation paper and interprets its limits. The current quantitative evaluation covers the YOLOv8l object-detection component. The paper reports a dataset of 350 training images and five validation images, a model trained for 100 epochs at image size 640 and batch size eight, and a confidence threshold of 0.60.

The detector achieved reported precision of 88.4%, mAP@0.5 of 61.3%, and an average inference time of 37.1 milliseconds per image. The small validation set is explicitly recognized as inadequate for strong generalizations claims. The results should be understood as an initial feasibility check, not as a representative estimate of performance on real examination scripts.

No quantitative OCR accuracy, connection detection accuracy, structure reconstruction accuracy, or expert-grading agreement is included in the supplied paper. These measures are discussed as pending tests rather than filled with estimates. This approach preserves the evidential boundary between reported measurements and planned experiments.

The custom YOLOv8l detector was trained to identify Attributes, Entities, Primary Keys, Relationships, and Weak Entities. The supplied paper reports the following aggregate results on a held-out validation set of five images. The results summaries the detector only, not the end-to-end ER-diagram grading pipeline.

The paper also mentions class-level figures and visualizations, but the extracted source does not provide a complete per-class numeric table. This thesis does not invent per-class precision, recall, or AP values.

The reported precision of 88.4% is higher than the reported mAP@0.5 of 61.3%. These metrics describe different aspects of detector behavior and should not be interpreted as interchangeable. Precision measures the share of predicted positives that match the evaluation definition, while mAP combines ranked precision-recall behavior across classes at the specified intersection-over-union threshold. A single aggregate number cannot identify which ER component classes are most difficult.

The average inference time of 37.1 milliseconds per image suggests that the detector stage completed quickly in the reported setup. It does not establish that a complete evaluation is fast enough for institutional use because crop preprocessing, calls to OCR.Space, line processing, structure reconstruction, and LLM inference add overhead. End-to-end latency should be measured across a representative batch, including service delays and failure retrieved.

The small validation set is the most important qualification on the reported values. With only five images, the sample may not represent the range of handwriting, diagram density, paper background, and image quality seen in actual examinations. The results are sensitive to which examples were selected and do not support a reliable estimate of performance on a new population of answer sheets.

The results are useful as a preliminary sanity check: the trained model produces detections, and the reported aggregate metrics show that the task is not entirely infeasible under the current data and setup. They are not sufficient to claim robust deployment readiness, parity with expert perception, or reliable grading of student scripts.

A stronger experiment should include a substantially larger independent test set and report per-class precision, recall, F1, AP, confidence intervals where appropriate, and representative error cases. The paper itself identifies independent testing as necessary before the figures can be treated as representative performance results.

Metric	Reported value	Interpretation
Training images	350	Number of images stated for the training split.
Validation images	5	Very small evaluation sample; limits confidence in generalizations.
Precision	88.4%	Reported proportion of positive detections judged correct under the experiment setup.
mAP@0.5	61.3%	Aggregate detection score at IoU 0.5 as reported by the paper.
Average inference time	37.1 ms/image	Detector-stage value; not a measured full-pipeline response time.
Image size	640 pixels	Configured image size reported for training.
Epochs / batch size	100 / 8	Reported training configuration.

Table 8.Preliminary object-detection results

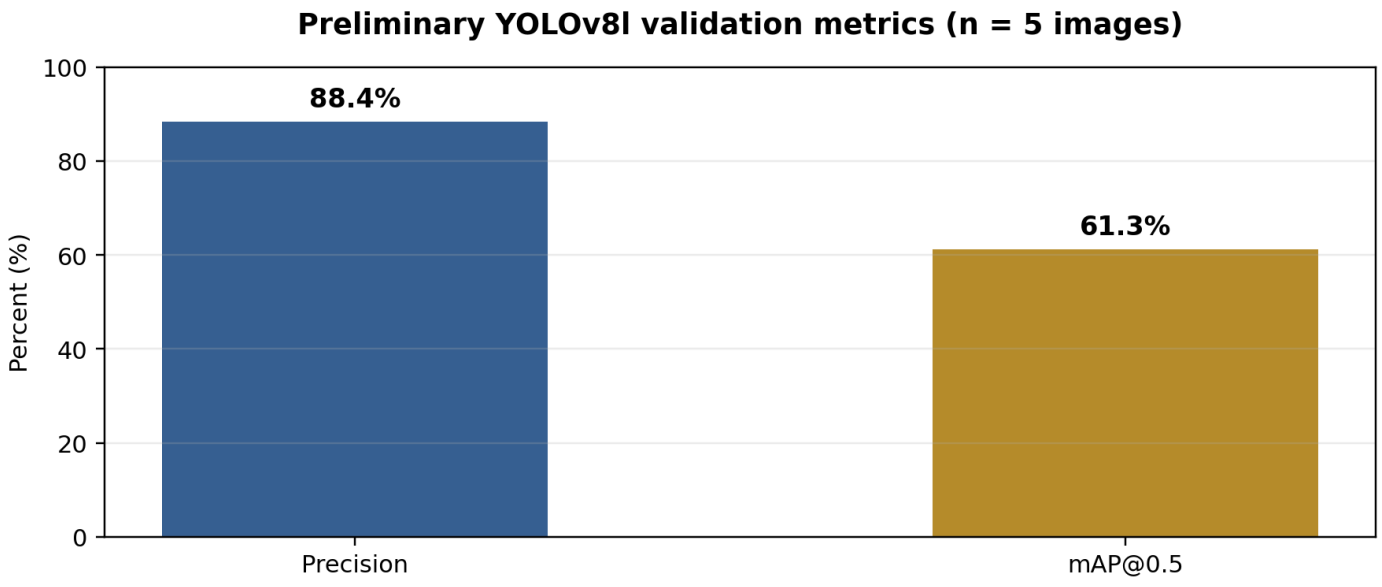


Figure 4.Preliminary object-detection metrics reported in the implementation paper.

The reported values establish that the custom detector was trained and evaluated on the stated small validation set. They do not establish that the model will generalize

to unseen examination cohorts, new handwriting styles, different paper formats, or diagrams with greater visual complexity. The five-image sample also means one or two detections can materially influence aggregate values.

6.2 Product and technical results

The implementation paper describes a set of outputs from the pipeline: annotated images, structured JSON, OCR results, detected connections, reconstructed diagram structure, criterion-level marks, and feedback. These outputs provide a route for tracing a decision from pixel-level detection to the final mark. The paper states that the output is intended to support both visual inspection and machine-readable use.

An annotated image can show whether the model selected the intended regions. Detection JSON preserves class names, coordinates, confidence, object identifier, and crop path. OCR records attach recovered labels to those objects. Connection records identify candidate line segments and object pairs. The reconstructed JSON then presents a consolidated model for rubric evaluation.

The existence of these artifact types supports the design claim that the pipeline is structured and inspectable. However, no independent evaluation of output completeness or accuracy is provided in the supplied result set. The schema shown in Appendix A is therefore an illustrative format for explaining the approach, not an extracted record from a particular student submission.

The paper describes a processing workflow that produces an annotated image, structured detection records, crop-level OCR output, candidate connections, a reconstructed ER-oriented JSON object, criterion-level marks, and feedback. These artefacts are valuable because they create an evidence trail from input image to proposed result. However, the available source material does not include a systematic count of successful end-to-end submissions, an error rate for the integrated pipeline, or a benchmark of total processing time. The technical result should be characterized as a described working/prototyped workflow with preliminary detector measurements, not as fully validated end-to-end automation.

6.3 User or stakeholder validation

No formal user or stakeholder validation is reported in the supplied proposal and implementation paper. The evidence does not include lecturer interviews, examiner acceptance ratings, student feedback, usability testing, time-on-task observations, or a comparison of user decisions with and without the system. Accordingly, this section does not report a user satisfaction percentage or claim that marking time was reduced.

A suitable next study would recruit lecturers or examiners with experience in database design assessment, subject to institutional approval. Participants could review a set of anonymized diagrams, compare manual and system-assisted workflows, identify incorrect detections and misleading feedback, and rate the usefulness of the intermediate artefacts. Measured outcomes should include time per answer, percentage of automated outputs accepted without correction, correction categories, perceived clarity, and confidence in the final grade. The study should be treated as a planned evaluation until data has actually been collected.

6.4 Integration results

Finding 1 - Object detection feasibility: the supplied detector was trained and evaluated, producing reported precision of 88.4% and mAP@0.5 of 61.3%. These results support further experimentation but remain preliminary because the validation set contains only five images.

Finding 2 - Modular architecture: the described pipeline separates detection, OCR, connection detection, reconstruction, and grading. This separation enables stage-level diagnosis. It also creates dependencies, so a mistake in an early stage can affect every later stage.

Finding 3 - Traceable intermediate outputs: the described outputs include images and structured data, which can support review and future audit. Their accuracy and completeness have not yet been independently quantified.

Finding 4 - Evaluation evidence is incomplete: OCR, structural reconstruction, connectivity, and grading agreement remain open evaluation areas in the supplied paper. Consequently, the research does not yet establish that automated marks agree reliably with experts or that the complete workflow reduces marking time.

Finding 5 - Human oversight remains necessary: known limitations such as OCR uncertainty, line crossing, nearby-object endpoint errors, and semantic variations mean that unclear cases should be routed for human review until more evidence exists.

The available source indicates that the components are arranged into a staged workflow and that structured outputs are created for downstream stages. Nevertheless, the provided results do not quantify the end-to-end execution success rate, the proportion of diagrams with valid reconstructed structures, the percentage of OCR outputs that match ground truth, or the agreement of automated scores with expert marks. These are crucial integration metrics because downstream quality depends on upstream detections, text extraction, and graph reconstruction.

A future integration result should be reported on a fixed set of complete diagrams with a known expected structure and an independently assigned human grade. For each image, the record should include the expected and predicted object set, expected and predicted connection set, reconstructed semantic structure, criterion scores, execution status, and processing duration. Error analysis should state at which stage the output first diverged from the annotated reference.

6.5 Summary of findings

Component / capability	Status supported by the supplied material	Evidence limitation
YOLOv8l detection of five ER classes	Implemented; preliminary metrics reported	Only five validation images.
Crop generation and OCR.Space processing	Described as implemented	OCR accuracy not reported.
Canny/Hough line detection	Described as implemented/prototyped	Connection quality not measured.
ER structure JSON reconstruction	Described as implemented/prototyped	Structural accuracy not measured.
Local LLM rubric grading	Grading module described	No expert-grade agreement result.
Background ruler removal	Requirement/design direction in proposal	No measured effectiveness reported.
Performance intelligence and automated viva	Part of the wider proposal	Not evaluated in this individual thesis.

Table 9. Implementation and validation status by component

The principal empirical finding is limited but clear: the detector achieved the reported preliminary values under the described setup. The principal system finding is that the proposed pipeline has a coherent staged design and produces interpretable intermediate artefacts according to the implementation paper. The principal evidence gap is the absence of broader independent and end-to-end validation.

7 DISCUSSION

7.1 Interpretation of findings

The work combines several components that have each been studied independently in computer vision and automated assessment. The primary contribution is the use of a single pipeline for image-based handwritten ER evaluation, rather than assuming the diagram is already digital and structured. The preliminary detector metrics demonstrate progress toward that objective, while the small validation sample sharply limits interpretation.

The largest technical risk is error propagation. A false detection changes what OCR sees; a wrong label changes the semantic representation; an incorrect connection changes the implied model; and the grading module may then assign a plausible but incorrect score. Therefore, improving detector precision alone will not necessarily provide acceptable final-grade agreement.

The nearest-object connection method is particularly sensitive to close component placement, crossing lines, and background rulings. OCR reliability can be affected by tiny or blurred labels and reliance on an external service. The LLM stage depends on the quality of the reconstructed input and its adherence to the rubric. Future work should measure each component independently, compare multiple valid diagrams, and evaluate final marks against expert annotation.

The broader proposal includes performance intelligence and automated viva assessment. These directions could eventually complement written diagram assessment, but they are outside the result scope of the current paper. Their implementation and validity would need separate designs, datasets, and evaluations.

The detector result should be interpreted as an initial feasibility signal, not as a reliable estimate of deployment performance. Precision alone does not reveal recall or missed objects, while mAP@0.5 is an aggregate metric that may hide differences between component classes. It is also possible for a detector to achieve acceptable object localisation but still fail to produce correct diagram semantics because OCR or line association is wrong. Therefore, the detector result is necessary evidence for the first stage but insufficient evidence for an automated marking claim.

A strength of the staged approach is diagnostic transparency. It can preserve information that allows a human examiner to locate where an error may have occurred. Its corresponding risk is error propagation: a missed or mislabeled object can contaminate OCR, graph reconstruction, and rubric evaluation. A production-quality design should treat upstream uncertainty as input to later decisions rather than discarding confidence values and presenting a deterministic score.

7.2 Comparison with literature baselines and existing solutions

Prior work on offline sketch parsing [6] and stroke reconstruction [7] establishes that static images are challenging because the original drawing sequence is missing. Arrow R-CNN [8] is particularly relevant because it studies handwritten diagram recognition with object detection and arrow-related structure. Automated feedback work such as Correia et al. [9] is relevant to educational use, while more recent LLM-based UML and ER assessment studies [4], [5] explore grading after diagrams have been converted to more structured representations.

The main distinction of the present component is its target input: diagram images drawn on examination papers rather than diagrams produced in a constrained digital editor. This changes the front-end problem because the system must find diagram objects and labels from pixels before semantic grading can begin. However, no fair numerical claim of superiority can be made from the current evidence. The detector was not evaluated on the datasets or annotation protocols used by the cited studies, and full-pipeline accuracy was not measured. Any future benchmark must use comparable classes, test conditions, and metrics or explicitly describe the differences.

The paper by Bouali et al. [4] describes comparing LLM-generated scores with teaching-assistant marks for digital UML diagrams, and Ramachandran et al. [5] address LLM-based ER/schema diagram assessment. These studies reinforce the need to validate the grading stage against human assessment. The current thesis does not borrow their reported grading agreement values as results for this implementation, because the systems, inputs, and evaluation sets differ.

7.3 Research contribution

The research contribution is a source-grounded design and preliminary technical evaluation of an image-first ER-diagram assessment pipeline. It connects object detection, crop-level text recognition, line-based connectivity estimation, structural reconstruction, and rubric-guided grading in a workflow oriented to handwritten examination answers. It also defines a component-level validation framework that separates the detection, OCR, connection, reconstruction, and grading stages so that error sources can be measured independently.

The contribution is preliminary because only detector metrics are reported. A fully demonstrated research contribution would require a larger labelled dataset, held-out tests, OCR and connectivity ground truth, expert-scored diagrams, comparison of machine and human scores, and documented false-positive/false-negative analysis. The present report identifies these requirements instead of implying they have already been completed.

7.4 Product contribution

At the product level, the described engine proposes a way to turn an image into reviewable data artefacts and criterion-level feedback. If validated, this could reduce the repetitive first-pass work involved in marking diagram answers and help lecturers apply rubric consistently. The product's differentiating target is handwritten, paper-based input rather than only diagrams authored with a digital modelling tool.

The current value should be described as a prototype capability and direction of integration. It has not been established that the product reduces marking time, improves fairness, lowers operational cost, or increases student learning outcomes. Those are hypotheses for subsequent product and stakeholder evaluation, not measured outcomes from the reported detector experiment.

7.5 Limitations and threats to validity

The largest internal-validity limitation is the small validation set of five images. The source material does not describe independent test data, repeated training runs, confidence intervals, or per-class metric values. The dataset may not represent the variety of handwriting, diagram complexity, paper rulings, image capture conditions,

or annotation practices in real examination settings. Without the original test predictions and annotations, the reported values cannot be independently recomputed from the supplied summary.

Construct validity is also a concern. Detector precision and mAP measure object detection, not the correctness of a full ER model or a grade. OCR text accuracy, graph-edge correctness, structure equivalence, and score agreement are separate constructs and require suitable ground-truth definitions. A line segment connected to the nearest object is not necessarily a semantically correct relationship. Likewise, a fluent feedback explanation is not evidence that the corresponding mark is justified.

External validity is limited by the focus on five diagram classes and a specific processing configuration. Generalization to different ER notations, UML diagrams, other diagram families, languages, or examination formats has not been established. Operational validity is unproven without load tests, service reliability tests, privacy review, or deployment monitoring. These threats mean the system should be used as a research prototype and not as the sole authority for high-stakes marking.

7.6 Implications

For research, the implication is that end-to-end diagram evaluation should be reported as a chain of measurable tasks. Improving detector precision alone will not necessarily improve the final mark if the line association or semantic reconstruction remains unreliable. The data and evaluation design should therefore include object labels, text transcriptions, connectivity annotations, complete semantic structures, and human marks for the same diagrams.

For education providers, a validated diagram assessment assistant could potentially support consistent initial feedback and help markers focus attention on ambiguous answers. But human accountability remains essential. Institutional adoption should begin with low-risk formative settings or a shadow evaluation in which system outputs are compared with normal marking without affecting grades. Only after suitable independent validation, fairness review, and governance should the role of automated suggestions be expanded.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

The proposal identifies universities, colleges, and other education providers as the main target market. Likely users include lecturers, examiners, examination administrators, and learning-support teams that need to process large numbers of diagram-based answers. The value proposition is faster initial processing, consistent criterion-level scoring, inspectable intermediate results, and support for structured feedback.

Possible revenue models include an annual institutional license, a semester-based subscription, or a tiered subscription based on enrolment or the number of exam scripts processed. Larger institutions may require integration with learning-management or examination systems, administrator roles, audit logs, and dedicated support. Smaller providers may prefer a limited-use hosted offering. These are potential models from the proposal, not validated market demand.

Costs identified in the proposal include AI and OCR services, cloud storage, hosting, connectivity, data collection, maintenance, and technical support. A commercial plan should separately estimate development cost, per-image compute cost, external OCR charges, language-model inference, security testing, customer support, and storage retention. No price should be finalized until cost per evaluation and willingness to pay have been studied.

The most direct value proposition is to assist lecturers with a repetitive and visually demanding first-pass assessment task. The product could present detected components, candidate relationships, a reconstructed structure, and rubric-level feedback in a single review workflow. The expected value lies in reducing manual inspection effort and making the evidence behind a suggested mark easier to examine; these benefits require validation with real users before being stated as demonstrated outcomes.

8.2 Market and competitive position

Before commercial release, the product would require a larger benchmark dataset, evidence of grading agreement, operational monitoring, support procedures, clear data-retention rules, and an interface that lets staff review errors. A human review

option is important for high-stakes assessment; an unreviewed score should not be treated as authoritative until the system has passed a suitable validation and governance process.

A hosted service could simplify updates and model management but raises privacy and residency questions for examination data. On-premises deployment could keep answer scripts inside an institution's environment but increases installation, hardware, and maintenance complexity. A hybrid architecture could host the application while running image processing inside the institution but would need a defined threat model and clear boundaries for external OCR or LLM calls.

The proposal's cost table estimates a total project budget of LKR 181,500, including software/services, cloud storage and hosting, connectivity, data-collection materials, and contingency. These are proposal estimates rather than audited costs or a commercial unit-economic model. Commercialization should replace them with current vendor quotes, actual usage measurements, and an estimate of support and operational labor.

Potential customers include universities, higher-education providers, vocational institutions, and organizations delivering technical training where diagram questions are assessed at scale. Competing approaches include manual marking, diagram editors with structured exports, general-purpose OCR and computer-vision APIs, rule-based assessment tools, and LLM-based grading systems that begin with textual or digitally structured diagrams. The distinctive proposed niche is paper-based handwritten ER-diagram input, but a durable commercial advantage depends on accuracy, reviewability, privacy, and integration with existing assessment workflows.

A product-market assessment should test whether examiners have a sufficiently large volume of diagram answers to justify adoption, what proportion of answers are suitable for automation, whether institution procurement permits cloud OCR or local model deployment, and what evidence is required for academic governance. Interviews and pilot deployments are needed to evaluate these questions. The current thesis does not report a market survey, competitor pricing study, or signed customer commitment.

8.3 Business or sustainability model

The proposal considers institutional annual licensing, semester-based subscriptions, and usage-based tiers according to the number of students or scripts processed. An on-premises deployment may suit institutions that require greater control over student data, whereas a hosted service may reduce local setup effort. These models are options for validation, not confirmed revenue arrangements.

Option	Potential benefit	Condition to validate
Institutional annual license	Predictable recurring revenue and institution-wide availability	Procurement fit, support expectations, and renewal intentions.
Semester subscription	Aligned with academic and examination cycles	Seasonal demand, account setup, and renewal rate.
Usage-based tier	Scales with the number of scripts processed	Predictable cost per image and peak-load handling.
On-premises deployment	Greater control of educational data	Hardware requirements and support burden.
API integration	Fits existing learning or examination systems	Interoperability, security, and change-management work.

Table 10.Commercialisation options

The proposal supplied a budget estimate of LKR 181,500. The values below are transcribed from the proposal and should be treated as project-level planning estimates from that document, not as current vendor quotations or a validated commercial cost model.

Cost category in proposal	Estimate (LKR)
LLM model / service	30,000
OCR model / service	20,000
Azure cloud storage - 10 months	30,000
Azure cloud hosting - 10 months	55,000
Wi-Fi / broadband - 10 months	25,000
Data collection materials and documentation	5,000
Contingency and miscellaneous	6,320

Cost category in proposal	Estimate (LKR)
Total stated in proposal	181,500

Table 11.Proposal-level budget estimate

A commercial cost model must separate fixed development costs from ongoing per-evaluation costs such as compute, OCR service calls, model inference, storage, monitoring, support, and security updates. Pricing should not be finalized until those costs are measured under realistic usage and willingness to pay is studied with institutions.

8.4 Scalability intellectual property and compliance

Scalability requires evaluation of batch sizes, concurrent image processing, inference resources, OCR API quotas, LLM throughput, storage retention, and queueing behavior. If the service processes examination batches, each image must remain isolated by evaluation identifier and errors must not allow one student's outputs to leak into another student's record. Load testing and monitoring are not reported in the supplied paper and remain prerequisites for operational scaling.

Intellectual-property review should cover ownership of collected or scanned examination images, the license terms of the object-detection framework and pretrained weights, the use conditions of the OCR service, the license for the locally hosted language model, and the right to use any annotation dataset commercially. The thesis does not establish legal clearance for these resources; permission and license review should be undertaken before distribution or sale.

Compliance planning should include institutional data policies, applicable privacy and records-management rules, security controls for credentials, deletion and retention policies, incident response, and accessibility of the review interface. Vendor/service terms should be examined before transmitting student work to an external API. These are responsible deployment requirements rather than evidence that compliance certification has already been obtained.

8.5 Contribution to the Sustainable Development Goals

The broader proposal aligns most directly with Sustainable Development Goal 4, Quality Education, because it explores ways to support timely feedback and consistent assessment. A usable assessment assistant could help educators handle repetitive marking, but a direct contribution to education quality cannot be claimed from detector precision alone. Evidence would be required that the tool supports accurate marking, accessible feedback, acceptable staff workload, and fair outcomes for students with varied handwriting or diagram styles.

The design also has sustainability considerations. Local processing may reduce some data-transfer dependencies but increase hardware and energy demand; hosted services may be easier to scale but incur recurring costs and raise data-transfer questions. Product sustainability should therefore be assessed through measured cost per answer, compute demand, infrastructure usage, data retention practices, and the operational lifetime of the service. These aspects are recommended evaluation topics, not current measured findings.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis examined the design and preliminary implementation of an image-based Diagram Evaluation Engine for handwritten ER diagrams on examination answer sheets. The motivation is the practical difficulty of consistently interpreting diagrammatic answers from paper, where component locations, handwriting, connectors, and background artefacts affect the evidence available to a marker.

The described pipeline combines YOLOv8l detection, OCR.Space OCR Engine 3, OpenCV Canny and probabilistic Hough line detection, structural reconstruction, and rubric-based grading through a locally hosted large language model. The detector is configured for five classes and was trained on 350 images with a five-image validation set. The reported preliminary measures are 88.4% precision, 61.3% mAP@0.5, and 37.1 milliseconds average inference time per image.

The central conclusion is that the implementation provides a useful feasibility foundation, but the supplied evidence does not yet establish reliable end-to-end marking. The detector's evaluation sample is too small for generalizations claims, and OCR, connectivity, reconstruction, and expert-grade agreement remain unmeasured in the reported results.

The first objective, detecting ER-diagram components, has preliminary quantitative evidence. The model was trained to identify Attributes, Entities, Primary Keys, Relationships, and Weak Entities, and the paper reports aggregate precision and mAP@0.5 on five validation images. This objective is partly supported at a proof-of-feasibility level, but a larger test set is required.

The second and third objectives, recovering labels and identifying connections, are described as implemented in the pipeline but lack reported accuracy measurements. These components need labelled test cases and separate evaluation. The fourth objective, constructing a structured ER representation, is supported by a described JSON-based workflow, but structural correctness is not quantified. The fifth objective, rubric-guided grading, is described architecturally, but no measured agreement with expert marks is available.

Accordingly, the work should be characterized as a prototype with a preliminary detection experiment. It should not be represented as a fully validated automated examiner. The research materials support the architecture and identify credible next steps, while further experiments are necessary to establish reliability, fairness, operational speed, and usefulness in an academic setting.

The central conclusion is that the research provides an initial technical basis for image-first assessment of handwritten ER diagrams, but the available evidence does not justify treating it as a validated autonomous marking system. The reported detector figures are promising enough to motivate a larger experiment, yet the five-image validation set is too small to support broad generalizations claims. OCR, connectivity, structural reconstruction, semantic grading, and stakeholder outcomes must be evaluated before the practical value of the full pipeline can be established.

9.2 Achievement of objectives

The objective of detecting ER-diagram components has preliminary quantitative evidence: a custom YOLOv8l model was configured for five classes and the paper reports 88.4% precision and 61.3% mAP@0.5, with 37.1 ms average detector inference time per image on the small validation sample. The objectives concerning OCR label recovery, connection identification, semantic structure reconstruction, and rubric-grading agreement are represented in the described processing architecture, but the supplied material does not provide quantitative validation for those stages. The objective of improving marking efficiency or consistency also remains unverified because no user study or expert-mark comparison is reported.

The objectives should therefore be categorized as follows: object detection has preliminary experiment evidence; downstream modules have implementation/prototype descriptions; end-to-end scoring quality and practical educational impact remain unmeasured. This classification is essential to keep the conclusion proportional to the study results.

9.3 Future directions

Future work should increase the size and diversity of the labelled dataset and separate training, validation, and test sets at the original-image level. Annotation guidelines

should be formalized and uncertain cases adjudicated. Model evaluation should include per-class precision and recall, AP, F1, error examples, and reproducible configuration details. Comparisons with smaller or alternative detectors should consider accuracy and computational cost.

OCR should be evaluated using character accuracy, word accuracy, exact label match, and failure rate. Connection detection should be evaluated using manually annotated object pairs and should account for lines that cross or overlap with paper rulings. Structure reconstruction should be tested on equivalent and non-equivalent ER diagrams. Final grades should be compared with expert marks using absolute error and criterion-level agreement.

Potential technical improvements include confidence-aware processing, better background-line handling, graph-based spatial reasoning, connector key point detection, and human correction of uncertain predictions. The wider proposal's text-answer analysis, performance intelligence, and automated viva functions require separate implementation and evaluation before they can be claimed as part of a delivered product.

A practical next phase should first expand the dataset and define annotation guidance for all five detector classes. Images should be split by original source or student where relevant to reduce leakage. After detector training, performance should be measured on a larger independent test set with per-class precision, recall, F1, AP, and mAP, along with representative false positives and false negatives. Repeated runs or uncertainty estimates should be added where suitable.

The second phase should establish ground truth for OCR labels and connections. OCR should be evaluated on both detected crops and the full expected set of components. Connectivity annotations should include the correct object pair for each relationship/connector, with explicit treatment of crossing lines, background rulings, and uncertain endpoints. Structural reconstruction should be checked against expert-reviewed semantic representations that permit equivalent layouts where valid.

The third phase should evaluate grading agreement using a rubric and independently assigned expert marks. It should report mean absolute error, criterion-level agreement,

error categories, and feedback quality, and test whether human review reduces the impact of uncertain inputs. User validation should observe marking time and correction rates in a controlled pilot. Only after those phases should the component be considered for broader institutional deployment or integration with the proposal's wider performance-intelligence and viva-assessment components.

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GLOSSARY

Term	Definition in the context of this thesis
Attribute	A property or field describing an entity in an ER model.
Bounding box	A rectangular coordinate region around a detected object.
Candidate connection	A hypothesized link between two diagram components inferred from line evidence.
Canny edge detection	An image-processing method used to identify intensity edges.
Confidence threshold	A minimum detector score used to accept a predicted object.
Connection reconstruction	Mapping detected line evidence to objects and representing candidate links.
Detection precision	The proportion of accepted detections that match evaluation ground truth.
Entity	An object or concept represented in an ER diagram.
Entity-Relationship diagram	A visual model of entities, attributes, keys, and relationships.
Ground truth	Manually verified annotations or answers used as an evaluation reference.
Held-out test set	Data excluded from training and model selection and used for final evaluation.
Intersection over Union (IoU)	The overlap ratio between a predicted and ground-truth bounding box.
Large Language Model (LLM)	A language model used here to compare a structured representation with a rubric.
Mean Average Precision (mAP)	The mean of Average Precision values across the evaluated classes at a stated IoU threshold.
Optical Character Recognition (OCR)	Technology that converts image text into machine-readable characters.
Primary key	An attribute or attribute set that uniquely identifies an entity instance.
Probabilistic Hough line detection	A technique for extracting candidate line segments from edge images.
Recall	The proportion of ground-truth objects recovered by the system.
Relationship	A semantic association between entities in an ER model.
Rubric	An explicit set of assessment criteria, descriptions, and mark limits.
Semantic equivalence	The condition in which different diagram presentations express the same underlying meaning.
SHA-256	A cryptographic hash function that produces a 256-bit digest.
Structural reconstruction	Conversion of detected components and candidate links into a machine-readable diagram model.
Validation set	Data used to assess behavior during development; not a substitute for an independent final test set.
Weak entity	An entity whose identification depends on an owner entity or identifying relationship.
YOLOv8l	The large YOLOv8 model variant used for the reported object-detection experiment.

APPENDICES

Appendix A Research instruments or extended results

This appendix collects supporting material for replication and future validation. The values explicitly labelled “reported” are transcribed from the supplied implementation paper. The proposed test questions and additional metrics are instruments for future execution and should not be read as evidence that the tests have passed.

Item	Value supported by supplied paper	Status / note
Detector	YOLOv8l	Reported
Training images	350	Reported
Validation images	5	Reported; insufficient for broad claims
Image size	640 pixels	Reported
Epochs	100	Reported
Batch size	8	Reported
Confidence threshold	0.60	Reported
Precision	88.4%	Preliminary result
mAP@0.5	61.3%	Preliminary result
Average inference time	37.1 ms/image	Detector stage; full-pipeline timing not reported
OCR accuracy	Not reported	Requires ground-truth transcriptions
Connection accuracy	Not reported	Requires labelled object-pair references
Grading agreement	Not reported	Requires expert marks on the same submissions

Suggested ground-truth collection form for each test diagram

- Image identifier and source permission status
- Image condition: scan/photo, resolution, skew, blur, shadows, and paper rulings
- Ground-truth objects: class, bounding-box coordinates, transcription, and key/entity semantics
- Ground-truth connections: source object ID, target object ID, connector type if used, and uncertainty notes
- Ground-truth ER structure: entities, attributes, key attributes, relationships, and constraints supported by the rubric
- Expert marks: criterion ID, maximum marks, awarded marks, short justification, and marker identifier
- Predicted outputs and stage-wise errors: detector, OCR, connection, structure, grading, and final review disposition

Metric definitions recommended for future reporting

- Precision = TP / (TP + FP), where TP and FP refer to detections matched or unmatched to the ground-truth class under the chosen matching threshold.
- Recall = TP / (TP + FN), where FN counts annotated objects that were not correctly detected.
- F1 = 2 × precision × recall / (precision + recall), with handling defined for zero denominators.
- Mean Absolute Error (MAE) for grading = average absolute difference between automated and expert marks over the same scoring units.
- Exact-label match = proportion of evaluated labels whose OCR transcription exactly matches the agreed reference string after a documented normalization policy.
- Connection precision/recall should be computed over annotated object pairs, not merely over geometric line segments.

Illustrative JSON structure (schema sketch; not an actual test result)


```
{
  "evaluation_id": "example-only",
  "image": "input-image.png",
  "objects": [
    {"id": "obj_1", "class": "Entity", "label": "Customer", "confidence": 0.91},
    {"id": "obj_2", "class": "Attribute", "label": "CustomerID", "confidence":
0.84}
  ],
  "connections": [
    {"source_id": "obj_1", "target_id": "obj_2", "status": "candidate",
"confidence": null}
  ],
  "grading": {"rubric_version": "example-only", "criteria": [], "total_mark":
null},
  "review_required": true
}
```


Appendix B Academic integrity and AI use disclosure

Draft disclosure for candidate review: “AI-based writing assistance was used to help organize the thesis structure, improve clarity, and prepare an initial draft from the project proposal and implementation paper. Technical claims, implementation descriptions, metrics, and references were checked against the supplied project materials and identified literature where available. AI assistance was not treated as experimental evidence, and no unreported performance values were intentionally added. The candidate remains responsible for verifying the technical accuracy of every

section, confirming citations, ensuring that the final text reflects their own work, and complying with the SLIIT policy on generative-AI use."

This statement is a draft and must be amended to match what the candidate actually used, the institutional policy in force, and any required declaration format. It should not be submitted unchanged if it does not truthfully describe the candidate's use of tools or if the Department requires a different disclosure form.

Academic-integrity checks before submission should include: verify each citation against the original source; quote only text that is explicitly quoted and cited; distinguish literature claims from the candidate's own findings; confirm that every reported result can be traced to an experiment record; and ensure the student contribution section is approved by the supervisor and project team.